

COURSE BOOK B. TECH. II YEAR

IT



KIET
GROUP OF INSTITUTIONS
Connecting Life with Learning



CURRICULUM STRUCTURE & SYLLABUS

Effective from the Session: 2025-26

1.1 Information Technology(IT)

B.Tech (IT) 3rd Sem

S No.	Course Category (AICTE)	Course Category (UGC)	SDG Mapping	BOS	Course Code	Course Name	Type	Academic Learning (AL)			Continuous Internal Examination (CIE)			End Sem Examination (ESE)	Total Marks	Total Credits
								L	T	P	MSE	CA	TOTAL			
1	PC	Major (Core)	SDG-4,9,11,16	IT	IT301L	Database Systems	L	3	0	0	60	15	75	75	150	3
2	PC	Major (Core)	SDG-4,8,9	CS	CS301L	Object Oriented Programming using Java	L	3	0	0	60	15	75	75	150	3
3	BS	Major (Core)	SDG-3,4,9	ASH	MA105L	Probability and Statistics	L	3	0	0	60	15	75	75	150	3
4	MC	Value Added	SDG-4,10,16	ASH	HS109L	Constitution of India	L	2	0	0	25	-	25	25	-	NC
5	HS	AEC	SDG-4,8,10	ASH	HS110L	Aptitude-1	L	1	0	0	-	25	25	25	50	1
6	HS	AEC	SDG-4,8,10	ASH	HS111L	Soft Skills Essential-1	L	1	0	0	-	25	25	25	-	NC
Blended																
7	PC	Major (Core)	SDG-4,8,9,12	CSE	CS302B	Advance Data Structure	B	3	0	2	80	20	100	100	200	4
8	PC	Major (Core)	SDG-4,9,11	IT	IT107B	Artificial Intelligence Essentials	B	3	0	2	80	20	100	100	200	4
9	PE	Major (Core)/SEC	-	-	-	Professional Elective-I	B	3	0	2	80	20	100	100	200	4
Lab/Practical																
10	PC	Major (Core)	SDG-4,9,11,16	IT	IT301P	Database Systems Lab	P	0	0	2	-	25	25	25	50	1
11	PC	Major (Core)	SDG-4,8,9	CS	CS301P	Object Oriented Programming using Java Lab	P	0	0	2	-	25	25	25	50	1
12	PW	Summer internship	SDG-4,10,11	CSIT	IT105P	Social Internship Assessment	P	0	0	0	-	50	50	-	50	1
Total Hours : 32 hrs.								22	0	10					1250	25

B. Tech (IT) 4th Sem

S No.	Course Category (AICTE)	Course Category (UGC)	SDG Mapping	BOS	Course Code	Course Name	Type	Academic Learning (AL)			Continuous Internal Examination (CIE)			End Sem Examination (ESE)	Total Marks	Total Credits
								L	T	P	MSE	CA	TOTAL			
1	PC	Major (core)	SDG-4,8,9,12	CSE	CS401L	Design and Analysis of Algorithms	L	3	0	0	60	15	75	75	150	3
2	ES	Major (core)	SDG-4,9,11	IT	IT302L	Computer Networks	L	3	0	0	60	15	75	75	150	3
3	PC	Major (Core)	SDG-4,8,9,12,16	CSE	CS206L	Operating System	L	3	0	0	60	15	75	75	150	3
4	MC	Value Added	SDG-3,4,16	ASH	HS112L	Universal Human Values	L	3	0	0	60	15	75	75	150	3
5	HS	AEC	SDG-4,8,10	ASH	HS113L	Aptitude-2	L	1	0	0	-	25	25	25	50	1
6	HS	AEC	SDG-4,8,10	ASH	HS114L	Soft Skills Essential-2	L	1	0	0	-	25	25	25	-	NC
Blended																
7	PC	Major (core)	SDG-4,8,9,11,12,16	CSE	CS208B	Web Technology	B	2	0	2	60	15	75	75	150	3
8	PE	Major (core)/SEC	-	-	-	Professional Elective-II	B	3	0	2	80	20	100	100	200	4
Lab/Practical																
9	PC	Major (core)	SDG-4,8,9,12	CSE	CS401P	Design and Analysis of Algorithms Lab	P	0	0	2	-	25	25	25	50	1
10	ES	Major (core)	SDG-4,9,11	IT	IT302P	Computer Networks Lab	P	0	0	2	-	25	25	25	50	1
11	PC	Major (Core)	SDG-4,8,9,12,16	CSE	CS206P	Operating System Lab	P	0	0	2	-	25	25	25	50	1
12	PC	Major (Core)	SDG-4,9,11	IT	IT205P	Data Engineering Architecture	P	0	0	4	-	50	50	50	100	2
Total Hours : 33 hrs.								19	0	14					1250	25

Professional Electives (PE)

Course Type (PE)	AI & ML	Cloud Operations (Powered by AWS)	Cloud Development (Powered by AWS)	Applied Apple Technologies: Swift, XCode & iOS Development (Powered by Apple)
SDG Mapping	SDG-4,9	SDG-4,9	SDG-4,9	SDG-4,9
BOS	CSE(AI) & CSE(AIML)	CSE(AI) & CSE(AIML)	CSE(AI) & CSE(AIML)	CSE(AI) & CSE(AIML)
PE I-(3rdSem)	Cloud Foundations (AI103E)	Cloud Foundations (AI103E)	Cloud Foundations (AI103E)	Cloud Foundations (AI103E)
BOS	IT	IT	IT	CSE
PE II-(4thSem)	Generative AI Foundations & Applications (IT315E)	Cloud Computing & Operations (IT311E)	Cloud Systems Development (IT318E)	Foundation of iOS App Development (CS321E)



1. Theory Courses Detail Syllabus

Course Code: IT301L				Course Name: Database Systems				L	T	P	C	
								3	0	0	3	
Pre-requisite: Concepts of any programming language												
Course Objectives:												
1. To develop a strong foundation in database management concepts.												
2. To equip students with practical skills in database design, normalization, transaction management, and recovery techniques.												
Course Outcome: After completion of the course, the student will be able to												
1. Acquire knowledge of database design methodology for real-life applications												
2. Design an information model using the concept of ER diagram												
3. Apply the concept of SQL on real-life databases												
4. Analyze the redundancy problem in the database and reduce it using normalization.												
5. Identify the broad range of database management issues including data integrity, security, and recovery transactions, as well as enforce entity integrity, referential integrity, key constraints, and domain constraints on the database.												
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)												
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	2	1	1	1	1	1	1	2
CO2	2	2	3	2	3	1	1	1	2	2	2	2
CO3	3	3	2	1	3	1	1	1	1	1	1	2
CO4	3	3	2	2	2	1	1	1	1	1	1	2
CO5	3	3	2	2	3	2	1	2	1	1	1	2
Unit 1	Introduction to Database System										09 hours	
Database System vs File System, Database System Concept and Architecture, Data Model Schema and Instances, Data Independence and its Types, Overall Database Structure.												
Entity Relationship Model: ER Model Concepts, Notation for ER Diagram, Mapping Constraints, Key attribute, Generalization, Aggregation, Reduction of an ER Diagrams to Tables.												
Use Case: Discuss one or two case studies like Banking System, and National Hockey League (NHL).												
Unit 2	Relational data Model										09 hours	
Relational Data Model Concepts, type of keys, Integrity Constraints- Entity integrity and referential integrity, Keys Constraints, Domain Constraints, Relational Algebra-Unary Relational Operations- SELECT and PROJECT, Binary Relational Operations-CROSS, JOIN and DIVISION, Queries in Relational Algebra.												
Database Implementation using SQL: Introduction to SQL, Characteristics of SQL, SQL Data Types, Basic Queries in SQL- create, select Insert, Delete and Update Statements, concepts of group by and having, order by, Sub Queries, Aggregate Functions, Joins, Unions, Intersection, Minus, Views.												
Note: Hands-on in the class/ Lab												
Unit 3	Database Design and Normalization										09 hours	
Functional Dependencies, Inference rules, Closure of attributes, FD equivalence and Minimal cover.												
Normalization: Normal forms, first, second, third normal forms, and BCNF. Lossless join decompositions, Dependency Preservation, Multi-valued Dependencies and Fourth Normal Form, Join Dependencies and Fifth Normal Form.												
Note: Discuss one or two case studies for the Normalization. e.g.												
1. In this a table would be given and students will be asked to normalize in a higher normal form Schema along with FDs set will be given and students will be asked to normalize in a higher normal form												
Unit 4	Transaction Processing										09 hours	
Transaction and its States, ACID property, Transaction Scheduling, Serializability of scheduling, Conflict, and View Serializability												
Concurrency Control Techniques: Concurrency Control, Locking Techniques for Concurrency Control, Two-phase locking techniques for concurrency control, Time Stamping Protocols for Concurrency Control												
Unit 5	Database Recovery Techniques										09 hours	
Recovery Concepts, Recoverability, Log Based Recovery, Checkpoints, Shadow Paging, The ARIES recovery, Deadlock Handling												
PL/SQL: Introduction, features, syntax, DDL within PL/SQL, DML in PL/SQL, Cursors, stored procedures, stored function, database triggers, indexing, Case Study-Microsoft Azure SQL.												
Note: Hands-on in the class/ Lab												

Total Lecture Hours					45 hours
Textbook:					
1. Elmasri, Navathe, “ Fundamentals of Database Systems”, Addison Wesley					
2. Korth, Silbertz, Sudarshan, “ Database Concepts”, McGraw Hill					
3. Date C J, “An Introduction to Database Systems”, Addison Wesley					
Reference Books:					
1. Bipin C. Desai, “An Introduction to Database Systems”, Gagotia Publications 8. Majumdar & Bhattacharya, “Database Management System”, TMH					
2. RAMAKRISHNAN “Database Management Systems”, McGraw Hill					
3. Rafe Colburn, Using SQL, PHI					
Mode of Evaluation					
MSE		CA			ESE
MSE1	MSE2	CA1	CA2	CA3 (ATTN)	
30	30	6	6	3	
60		15			
					Total
					75
					150

Course Code: CS206L		Course Name: Operating System					L	T	P	C		
							3	0	0	3		
Pre-requisite: Basic knowledge on Computer System and system memory, Computer Organization and Logic Design (COLD)												
Course Objectives:												
This course aims to provide a comprehensive understanding of operating systems, their components, structures, and functionalities. It covers fundamental concepts such as process management, memory management, file systems, and concurrency. The course also emphasizes practical aspects, such as system calls, Linux commands, shell scripting, and scheduling algorithms. By the end of this course, students will have the knowledge and skills to understand, analyze, and apply operating system principles in real-world scenarios.												
Course Outcome: After completion of the course, the student will be able to												
1. Understand the need, evolution and design issues of various categories of operating systems.												
2. Apply different CPU scheduling algorithms and deadlock handling methods.												
3. Analyze the principles of concurrency control and process synchronization problem.												
4. Analyze various memory management techniques for efficient memory allocation.												
5. Apply the concept of various I/O management, Disk scheduling and file system.												
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)												
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	1	1	-	-	-	-	-	2
CO2	3	3	3	3	3	1	-	-	-	-	-	2
CO3	3	3	2	3	3	2	-	-	-	-	-	2
CO4	3	3	2	3	3	2	-	-	-	1	-	2
CO5	3	3	2	2	2	2	-	-	-	1	-	2
Unit 1	Introduction of Operating System										09 hours	
Introduction: Operating system Components and its services, Classification of Operating systems- Batch system, Time sharing, Real Time System, Multiprocessor Systems, Multiuser Systems, Multiprocess Systems, Multithreaded Systems, Operating System Structure- Layered structure, Reentrant Kernels, Monolithic and Microkernel Systems. System Calls, Elementary Linux commands and Shell Scripting.												
Unit 2	Process Scheduling and Resource Management										09 hours	
Introduction to Process: Process States, State Transition Diagram, Schedulers, Process Control Block (PCB), Threads and their management, CPU Scheduling: Concepts, Performance Criteria, Scheduling Algorithms. Deadlock: System model, Deadlock characterization, Prevention, Avoidance and detection, Recovery from deadlock.												
Unit 3	Concurrent Processes										09 hours	
Concurrent Processes: Principle of Concurrency, Critical Section Problem, Mutual Exclusion, Dekker’s solution, Peterson’s												

solution, Semaphores, Monitors, Test and Set operation; Classical Problem in Concurrency- Producer / Consumer Problem, Reader Writer Problem, Dining Philosopher Problem, Sleeping Barber Problem; Inter Process Communication models (IPC).							
Unit 4		Memory Management				09 hours	
Memory Management: Basic bare machine, Resident monitor, Multiprogramming with fixed partitions and variable partitions, Protection schemes, Paging, Segmentation, Paged segmentation, Virtual memory concepts, Demand paging, Performance of demand paging, Page replacement algorithms, Thrashing.							
Unit 5		I/O Management and Disk Scheduling:				09 hours	
File Systems and I/O Management of Windows and Linux.							
Disk Scheduling: Disk storage structure and disk scheduling algorithms, RAID.							
Case Study: Introduction to Android and Mac Operating System, The Evolution of Mobile Operating Systems: iOS vs. Android							
					Total Lecture Hours	45 hours	
Text Book:							
1. Silberschatz, Galvin and Gagne, “Operating Systems Concepts”, Wiley							
2. Harvey M Dietel, “ An Introduction to Operating System”, Pearson Education							
3. William Stallings, “Operating Systems: Internals and Design Principles ”, 6 th Edition, Pearson Education							
Reference Book:							
1. Sibsankar Halder and Alex A Aravind, “Operating Systems”, Pearson Education							
2. D M Dhamdhare, “Operating Systems : A Concept based Approach”, 2 nd Edition, TMH.							
3. Andrew S. Tanenbaum and Herbert Bros, Modern Operating Systems (4thEdition), Pearson							
Mode of Evaluation							
MSE		CA			ESE	Total	
MSE1	MSE2	CA1	CA2	CA3 (ATTN)			
30	30	6	6	3			
60		15					

Course Code: CS301L	Course Name: Object Oriented Programming using Java								L	T	P	C
									3	0	0	3
Pre-requisite: NA												
Course Objectives:												
1. To familiarize students with the basic and advance Java Programming Language.												
2. To learn modern tools to develop java-based web applications.												
Course Outcome: After completion of the course, the student will be able to												
1. Implement core Java concepts that model real world entities.												
2. Develop programs based on new java features.												
3. Apply a collection framework to build modular java programs.												
4. Construct dynamic web pages using JDBC, Servlets and JSP.												
5. Implement web and RESTful Web Services with Spring Boot using Spring Framework concepts.												
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)												
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	2	2	2	-	-	-	-	-	-	2
CO2	2	3	2	2	2	-	-	-	-	-	-	2
CO3	2	3	2	2	2	-	-	-	-	-	-	3
CO4	2	2	2	2	3	-	-	-	-	-	2	3
CO5	2	3	3	3	2	-	-	-	-	-	2	3
Unit 1	Java Basics										09 hours	
Introduction to Java, JVM, JRE, Java Environment, Java Source File Structure, and Compilation. Defining Classes in Java, Constructors, Methods, Access Specifiers, Static Members, Final Members, Class, Object, Abstraction, Inheritance, Encapsulation, Polymorphism, Exceptions: Use of try, catch, finally, throw, throws, In-built and User Defined Exceptions, Checked and Un-Checked Exceptions.												

Unit 2	Java Features	09 hours					
Thread, Thread Life Cycle, Creating Threads, Thread Priorities; Advance Features: Functional Interfaces, Lambda Expression, Method References, Stream API, Default Methods, Static Method, For Each Method, Try-with-resources, Type Annotations, Repeating Annotations, Java Module System, Diamond Syntax with Inner Anonymous Class, Local Variable Type Inference, Switch Expressions, Yield Keyword, Records, Sealed Classes.							
Unit 3	Java Collections	09 hours					
Collection in Java, Collection Framework in Java, Hierarchy of Collection Framework, Iterator Interface, Collection Interface List Interface, ArrayList, LinkedList, Vector, Stack, Queue Interface, Set Interface, HashSet, LinkedHashSet, Sorted Set Interface, TreeSet, Map Interface, HashMap Class, Linked HashMap Class, TreeMap Class, Hashtable Class, Sorting, Comparable Interface, Comparator Interface, Properties Class in Java.							
Unit 4	Advance Java	09 hours					
Java Database Connectivity (JDBC): Merging Data from Multiple Tables: Joining, Manipulating, Databases with JDBC, Prepared Statements, Transaction Processing, Stored Procedures, Servlet Overview and Architecture, Interface Servlet and the Servlet Life Cycle, Handling HTTP get Requests, Handling HTTP post Requests, Redirecting Requests to Other Resources, Session Tracking, Cookies, Session Tracking with Http Session, Java Server Pages, Implicit Objects, Scripting, Standard Actions, Directives.							
Unit 5	Spring and Spring boot	09 hours					
MVC, Spring Core Basics, Spring Dependency Injection concepts, Spring Inversion of Control, AOP, Bean Scopes- Singleton, Prototype, Request, Session, Application, Web Socket, Auto wiring, Annotations, Life Cycle Call backs, Bean Configuration styles; Spring Boot Build Systems, Spring Boot Code Structure, Spring Boot Runners, Logger, BUILDING RESTFUL WEB SERVICES, Rest Controller, Request Mapping, Request Body, Path, Variable, Request Parameter, GET, POST, PUT, DELETE APIs, Build Web Applications.							
Total Lecture Hours		45 hours					
Textbook: 1. Herbert Schildt, "Java The complete reference", McGraw Hill Education 2. Steven Holzner, “Java Black Book”, Dreamtech. 3. Balagurusamy E, “Programming in Java”, McGraw Hill 4. Java: A Beginner’s Guide by Herbert Schildt, Oracle Press 5. Greg L. Turnquist “Learning Spring Boot 2.0 - Second Edition”, Packt Publication							
Reference Books: 1. Kathey Sierra, “Head First Java”, O’Reilly 2. AJ Henley Jr (Author), Dave Wolf, “Introduction to Java Spring Boot: Learning by Coding”, Independently Published 3. Craig Walls, “Spring Boot in Action” Manning Publication							
Mode of Evaluation							
MSE		CA	ESE	Total			
MSE1	MSE2	CA1				CA2	CA3 (ATT)
30	30	6				6	3
60		15					
		75		150			

Course Code: MA105L	Course Name: Probability & Statistics								L	T	P	C
									3	0	0	3
Pre-requisite: X+2												
Course Objectives:												
1. To familiarize the graduate engineers with the concept of Statistics and Probability. 2. It aims to analyze the practical/ real life problems and solve them in scientific manner.												
Course Outcome: After completion of the course, the student will be able to												
1. Employ the concept of measure central tendency and regression analysis. 2. Apply knowledge of probability on distribution function. 3. Apply the concept of probability density function and normal distribution. 4. Apply the concept of random variable and time series. 5. Employ the knowledge of hypothesis by means of Chi-square and ANOVA test.												
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)												
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12

CO1	2	2	2	-	-	-	-	1	-	-	-	1
CO2	2	2	2	-	-	-	-	1	-	-	-	1
CO3	2	2	1	-	-	-	-	1	-	-	-	1
CO4	2	2	1	-	-	-	-	1	-	-	-	1
CO5	2	2	2	-	-	-	-	1	-	-	-	1
Unit 1	Basic Statistics										09 hours	
Introduction to Descriptive Statistics, Measure of Central Tendency, Histogram in sampling, Method of least square (basic concept), Fitting of Straight line and exponential curve, Correlation, Rank correlation and Regression Analysis.												
Unit 2	Probability I										09 hours	
Probability, Law of total Probability, Conditional Probability, Baye’s Theorem, Discrete Random Variable, Probability Mass function. Binomial Distribution, Poisson Distribution., Introduction to confusion matrix.												
Unit 3	Probability II										09 hours	
Continuous Random Variable, Probability density function, Properties of Probability density function, Expectation and variance, Normal Distribution and its applications.												
Unit 4	Bivariate Random Variable and Time Series										09 hours	
Introduction to two dimensional random variable, Joint probability density function and its properties, Marginal probability distribution. Introduction to Time series, component of time series, Measure of trend (Graphic method, method of Averages)												
Unit 5	Sampling Theory										09 hours	
Introduction to Inferential Statistics, Testing of Hypothesis: Introduction, Sampling Theory (Small and Large), Hypothesis, Null hypothesis, Alternative hypothesis, Testing a Hypothesis, Level of significance, Confidence limits, t-test, Chi-square test, one way analysis of variance (ANOVA).												
										Total Lecture Hours		45 hours
Textbook:												
1. B. V. Ramana, Higher Engineering Mathematics, McGraw-Hill Publishing Company Ltd., 2017												
2. B. S. Grewal, Higher Engineering Mathematics, Khanna Publisher, 2017.												
3. R K. Jain & S R K. Iyenger, Advance Engineering Mathematics, Narosa Publishing House 2002.												
4. S. C. Gupta & V. K. Kapoor, Fundamental of Mathematical Statistics, Sultan Chand & Sons.												
Reference Books:												
1. Seymour Lipschutz, John Schiller, Introduction to Probability and Statistics, McGraw Hill												
2. Peter V. O’Neil, Advance Engineering Mathematics, Thomson (Cengage) Learning, 2007.												
3. TKV Iyenger, B. Krishna Gandhi, S. Ranganatham, MVS N Prasad, Probability and Statistics (S. Chand Publishing House).												
4. E. Kreyszig, Advance Engineering Mathematics, John Wiley & Sons, 2005.												
Mode of Evaluation												
MSE		CA			ESE	Total						
MSE1	MSE2	CA1	CA2	CA4 (ATT)								
30	30	6	6	3								
60		15										

Course Code: HS109L	Course Name: Constitution of India					L	T	P	C
						2	0	0	NC
Pre-requisite: NA									
Course Objectives:									
1. To acquaint the students with legacies of constitutional development in India and help those to understand the most diversified legal document of India and philosophy behind it.									
2. To make students aware of the theoretical and functional aspects of the Indian Parliamentary System.									
3. To channelize students' thinking towards basic understanding of the legal concepts and its implications for engineers.									
4. To learn procedure and effects of emergency, composition and activities of election commission and amendment procedure.									
Course Outcome: After completion of the course, the student will be able to									
1. Understand basic features and modalities about Indian constitution.									
2. Clarify the functioning of Indian parliamentary system at the center and state level.									

3. Understand the aspects of Indian Legal System and its related bodies.
4. Apply different laws and regulations related to engineering practices.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	-	1	2		-	-	2
CO2	-	-	-	-	-	-	1	1	1	-	-	2
CO3	-	-	-	-	-	-	1	1	1	-	1	2
CO4	-	-	-	-	-	-	1	2	1	1	1	2

Unit 1 Basic Information about Indian Constitution**08 hours**

Meaning of the constitution law and constitutionalism, Historical Background of the Constituent Assembly, Government of India Act of 1935 and Indian Independence Act of 1947, Enforcement of the Constitution, Indian Constitution and its Salient Features, The Preamble of the Constitution, Fundamental Rights, Fundamental Duties, Directive Principles of State Policy, Parliamentary System, Federal System, Centre-State Relations, Amendment of the Constitutional Powers and Procedure, The historical perspectives of the constitutional amendments in India, Emergency Provisions: National Emergency, President Rule, Financial Emergency, and Local Self Government – Constitutional Scheme in India.

Unit 2 Union Executive and State Executive**08 hours**

Powers of Indian Parliament Functions of Rajya Sabha, Functions of Lok Sabha, Powers and Functions of the President, Comparison of powers of Indian President with the United States, Powers and Functions of the Prime Minister, Judiciary – The Independence of the Supreme Court, Appointment of Judges, Judicial Review, Public Interest Litigation, Judicial Activism, Lok Pal, Lok Ayukta, The Lokpal and Lok ayuktas Act 2013, State Executives – Powers and Functions of the Governor, Powers and Functions of the Chief Minister, Functions of State Cabinet, Functions of State Legislature, Functions of High Court and Subordinate Courts.

Unit 3 Basic Information about Legal System**07 hours**

The Legal System: Sources of Law and the Court Structure: Enacted law -Acts of Parliament are of primary legislation, Common Law or Case law, Principles taken from decisions of judges constitute binding legal rules. The Court System in India and Foreign Courtiers (District Court, District Consumer Forum, Tribunals, High Courts, Supreme Court). Arbitration: As an alternative to resolving disputes in the normal courts, parties who are in dispute can agree that this will instead be referred to arbitration. Contract law, Tort, Law at workplace.

Unit 4 Election provisions, Emergency provisions, Amendment of the constitution**07 hours**

Election Commission of India-composition, powers and functions and electoral process. Types of emergency-grounds, procedure, duration and effects. Amendment of the constitution- meaning, procedure and limitations

Total Lecture Hours**30 hours****Textbook**

1. Brij Kishore Sharma: *Introduction to the Indian Constitution*, 8th Edition, PHI Learning Pvt. Ltd.
2. Granville Austin: *The Indian Constitution: Cornerstone of a Nation (Classic Reissue)*, Oxford University Press.
3. S.G Subramanian: *Indian Constitution and Indian Polity*, 2nd Edition, Pearson Education 2020.
4. Subhash C. Kashyap: *Our Constitution: An Introduction to India's Constitution and constitutional Law*, NBT, 2018.
5. Madhav Khosla: *The Indian Constitution*, Oxford University Press.
6. PM Bakshi: *The Constitution of India*, Latest Edition, Universal Law Publishing.
7. V.K. Ahuja: *Law Relating to Intellectual Property Rights* (2007)
8. Suresh T. Viswanathan: *The Indian Cyber Laws*, Bharat Law House, New Delhi-88
9. P. Narayan: *Intellectual Property Law*, Eastern Law House, New Delhi
10. Executive programme study material Company Law, Module II, by ICSI (The Institute of Companies Secretaries of India) (Only relevant sections i.e., Study 1, 4 and 36). <https://www.icsi.edu/media/webmodules/publications/Company%20Law.pdf>
11. Handbook on e-Governance Project Lifecycle, Department of Electronics & Information Technology, Government of India, https://www.meity.gov.in/writereaddata/files/e-Governance_Project_Lifecycle_Participant_Handbook-5Day_CourseV1_20412.pdf
12. Companies Act, 2013 Key highlights and analysis by PWC. <https://www.pwc.in/assets/pdfs/publications/2013/companies-act-2013-key-highlights-and-analysis.pdf>

Reference Books

1. Keshavanand Bharati V. State of Kerala, AIR 1973 SC 1461.
2. Maneka Gandhi V. Union of India AIR, 1978 SC 597.
3. S.R. Bammai V. Union of India, AIR 1994 SC 1918.



4. Kuldip Nayyar V. Union of India, AIR 2006 SC312.
 5. A.D.M. Jabalpur V. ShivkantShakla, AIR 1976 SC1207.
 6. Remshwar Prasad V. Union of India, AIR 2006 SC980.
 7. Keshav Singh in re, AIR 1965 SC 745.
 8. Union of India V. Talsiram, AIR 1985 SC 1416.
 9. Atiabari Tea Estate Co.V. State of Assam, AIR 1961SC232.
 10. SBP & Co. Vs. Patel Engg. Ltd. 2005 (8) SCC 618.
 11. Krishna Bhagya Jala Nigam Ltd. Vs. G. Arischandra Reddy (2007) 2 SCC 720.
 12. Oil & Natural Gas Corporation Vs. Saw Pipes Ltd. 2003 (4) SCALE 92 – 185.
 13. Contemporary Newer case studies can be developed using AI tools
- ** (Other relevant case studies can be consulted by the teacher as per the topic). Prescribed Legislations:
1. Information Technology Act, 2000 with latest amendments. **Compare this with GDPR of Europe**
 2. RTI Act 2005 with latest amendments.
 3. Information Technology Rules, 2000
 4. Cyber Regulation Appellate Tribunal Rules, 2000

Suggested aid for Students and Pedagogic purpose

- RSTV debates on corporate law, IPR and patent issues
- NPTEL lectures on IPR and patent rights

Episodes of 10 -part mini TV series “Samvidhan: The Making of Constitution of India” by RSTV.

Mode of Evaluation

MSE		CA			ESE	Total	
MSE	MSE2	CA1	CA2	CA4 (ATT)			
-	25	-	-	-	25	NC	
25		-					

Course Code: HS110L	Course Name: Aptitude-1							L	T	P	C	
								1	0	0	1	
Pre-requisite: NA												
Course Objectives:												
1. To provide adequate exposure to the students regarding the use of aptitude tests in the recruitment process and competitive examinations.												
2. To improve the logical & numerical ability of the students												
Course Outcome: After completion of the course, the student will be able to												
1. Illustrate their comprehension by solving the given problems												
2. Apply the learned concepts to new problems and solve them aptly.												
3. Make use of their thought process to interpret and draw inferences from the given data to reach logical conclusions.												
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)												
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	-	1	-	1		-	-	-	-	1
CO2	1	1	-	1	-	2		-	-	-	-	1
CO3	1	1	-	1	-	1		-	-	-	-	2
Unit 1	Series , Coding and Decoding										04 hours	
Importance and overview of Quantitative Aptitude and Logical Reasoning, Number Series, Letter Series, Analogies , Coding and Decoding.												
Unit 2	Data Arrangement										04 hours	
Ranking and Order, Direction Sense, Linear and Circular sitting arrangement.												
Unit 3	Blood Relation and Puzzles										03 hours	
Basic concepts, definition and terminology related to blood relationships, Conversation based blood relationships, Family Tree based problems, Coded relationships and related puzzles.												
Unit 4	Critical and Non Verbal Reasoning										04 hours	
Statement arguments, course of action, classification and grouping of images, Figure series, Mirror image, Water image, Paper cutting, Paper folding, Embedded figures.												



Course Code: HS111L				Course Name: Soft Skills Essentials-1				L	T	P	C	
								1	0	0	NC	
Pre-requisite: - Students should have foundational knowledge of grammar, vocabulary, and sentence structure to participate effectively in tasks like extempore, scenario writing after studying Communication skills subject in first year. - Prior exposure to basic communication concepts (like verbal/non-verbal communication and listening skills) helps students to enhance persuasion, negotiation, and professional etiquette.												
Course Objectives:												
To develop students' communication, presentation, and interpersonal skills through interactive activities, elevating confidence and professionalism for academic and workplace success												
Course Outcome: After completion of the course, the student will be able to												
- Demonstrate improved self-awareness and communication skills through structured presentations and vocabulary-building activities. - Apply effective verbal communication techniques, including pronunciation and elevator pitch delivery, to express ideas clearly and confidently. - Exhibit professional behaviour, grooming, and teamwork skills in group discussions, interviews, and workplace-related role plays.												
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)												
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	-	-	-	2	3	-	1
CO2	-	-	-	-	-	-	-	-	2	3	-	2
CO3	-	-	-	-	-	-	-	-	2	3	-	2
Unit 1	Foundation of Communication and Self-Awareness										05 hours	
British Council-English Score Test, Team Presentations on Change Management Models, Presentations on Personality Profiling for professional growth												
Unit 2	Verbal Communication and Clarity										04 hours	
Pronunciation Drill 1 & 2, Elevator Pitch Practice Session 1 & 2												
Unit 3	Professionalism and Workplace Readiness										06 hours	
Professional Grooming and Etiquette, Group Discussion (General Topics), Panel Discussion on workplace scenarios using caselets												
Total Lecture Hours										15 hours		
Useful Resources: - www.mindtools.com - https://englishonline.britishcouncil.org/ - www.toastmasters.org - https://www.futurelearn.com/ - English Score Test - Duo Lingo Test												
Mode of Evaluation												
MSE	CA				ESE	Total						

MSE1	MSE2	CA1	CA2	CA3(ATT)		
-	-	10	10	5		
-		25			25	50

Course Code: CS302B				Course Name: Advance Data Structure						L	T	P	C
										3	0	2	4
Pre-requisite: The course requires a background in mathematics and strong programming skills, along with fundamental knowledge of data structures and algorithms.													
Course Objectives:													
1. This course develops a deep understanding of advanced data structures for efficient problem-solving.													
2. Students will analyze, implement, and apply complex structures like trees, heaps, graphs, and hashing.													
Course Outcome: After completion of the course, the student will be able to													
1. Analyze algorithms using asymptotic notations and apply the Divide & Conquer technique for efficient problem-solving.													
2. Implement hashing techniques with collision resolution strategies for optimized searching and storage.													
3. Construct and manipulate tree-based data structures, including BST, AVL, Red-Black, B-Trees, and Huffman coding.													
4. Apply graph representations and traversal techniques like BFS, DFS, Connected Components, and Topological Sorting.													
5. Utilize advanced data structures such as Binomial Heaps, Tries, Skip Lists, and Splay Trees for efficient data processing.													
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)													
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	
CO1	3	3	3	2	-	-	-	1	-	-	1	2	
CO2	3	3	3	2	-	-	-	1	-	-	1	2	
CO3	3	3	3	2	-	-	-	1	-	-	1	2	
CO4	3	3	3	2	-	-	-	1	-	-	1	2	
CO5	3	3	3	2	-	-	-	1	-	-	1	2	
Unit 1	Introduction & Divide and Conquer Technique										15 hours		
Introduction: Algorithms, Analyzing algorithms - Orders of Magnitude (Asymptotic notations), Growth of functions (constant, logarithmic, linear, polynomial, exponential). Divide & Conquer Technique: Binary search, Merge sort, Quick sort, Randomized quick sort, Efficiency analysis of sorting.													
Problem solving:													
1. Write a program to implement the Binary Search algorithm using Divide and Conquer technique on a sorted array. Accept an element from the user and search for it in the array. Display the number of comparisons made.													
2. Given an array of integers nums sorted in non-decreasing order, and an integer target, find the starting and ending position of the target value. If the target is not found in the array, return [-1, -1]. Your solution must run in O (log n) time complexity.													
3. Write a program to implement Merge Sort using Divide and Conquer. Accept n elements from the user, sort them, and display the sorted output along with the number of recursive calls and comparisons made.													
4. Given an array of integers, count the number of inversions in the array. An inversion is a pair of elements (i, j) such that:													
• i < j and													
• arr[i] > arr[j]													
Your task is to write a program that returns the total number of such inversion pairs in the array.													
5. Write a program to implement Quick Sort using the last element as the pivot. Accept an unsorted array, sort it, and count the number of comparisons and recursive calls.													
6. Write a program to implement Randomized Quick Sort where the pivot is chosen randomly. Compare the number of comparisons and execution time with the regular Quick Sort.													
7. Given an integer array nums, find the contiguous subarray (containing at least one number) which has the largest sum, and return its sum using the Divide and Conquer approach.													
Given an integer array nums and an integer k, return the kth largest element in the array. Note that it is the kth largest element in sorted order, not the kth distinct element.													
Unit 2	Hashing										15 hours		
Hashing & Searching Techniques: Introduction, Hash table, Hash function, Collision resolution technique: Open hashing (Separate Chaining), Closed Hashing - Linear Probing, Quadratic probing, Double Hashing.													
Problem solving:													

- Write a program to implement a hash table using the open hashing (separate chaining) method. Use an array of linked lists to handle collisions. Support insertion, deletion, and search operations. Test your implementation by inserting multiple keys and observing how collisions are handled using chaining. - Given an array of integers, find the frequency of each distinct element using hashing and return a map of element to its count. - Given an array and an integer k, check whether the array contains duplicate elements within a distance k from each other using hashing. - Given an array and a value sum, determine whether any two numbers in the array add up to the sum using a hash set for efficient search. - Given an array and a window size k, print the count of distinct elements in every contiguous window of size k using hashing. - Given an integer array $nums$ and an integer k, return the k most frequent elements. You may return the answer in any order. - Given an array of strings $strs$, group the anagrams together. You can return the answer in any order. Two strings are anagrams of each other if they contain the same characters in the same counts, but possibly in a different order. - Given an array of integers $nums$ and an integer $target$, return indices of the two numbers such that they add up to the target.		
Unit 3	Trees	15 hours
Tree: Binary Trees, Binary Search Trees (BST), Threaded Binary Trees, Huffman coding, Balanced Trees: AVL Tree & its operation (insertion, deletion). Problem solving: - Write a program to implement a binary tree with basic operations such as insertion (level-order), in-order, pre-order, and post-order traversals. Test the tree with a set of integers and display the output of each traversal method. - Implement a binary search tree (BST) with functions to insert, delete, and search a node. Perform and display in-order traversal after each operation to show the structure of the BST. - Write a program to build a Huffman tree for a given set of characters and their frequencies, and generate Huffman codes for each character. Display the codes and the structure of the tree used in encoding. - Create a program to implement an AVL Tree that supports insertion of elements. Display the tree structure (in-order traversal) after each insertion and show the rebalancing operations (rotations) performed. - Given the root of a binary tree, find its maximum depth (also called height). The maximum depth is the number of nodes along the longest path from the root node down to the farthest leaf node. - Given two integer arrays, preorder and inorder, where: preorder is the Preorder Traversal of a binary tree, inorder is the Inorder Traversal of the same tree, Construct and return the binary tree. - Given the root of a binary tree, return the length of the diameter of the tree. The diameter of a binary tree is the longest path between any two nodes in the tree. This path may or may not pass through the root. Note: The length of a path between two nodes is measured by the number of edges between them. - Given a binary tree and two nodes p and q, find their lowest common ancestor (LCA). The LCA of two nodes p and q is the lowest node in the tree that has both p and q as descendants (where a node can be a descendant of itself).		
Unit 4	Graph	15 hours
Graphs: Terminology, Sequential and linked Representations of Graphs: Adjacency Matrices, Adjacency List, Adjacency Multi list, Graph Traversal: Depth First Search and Breadth First Search, Connected Component, Topological sort. Problem solving: - Implement a graph using an adjacency matrix and provide functions to: - Add/remove vertices and edges. - Check if an edge exists between two vertices. Print the matrix. - Implement a graph using an adjacency list (linked lists or dynamic arrays) and support: - Edge insertion/deletion. - Display neighbors of a vertex. - Implement DFS to traverse a graph from a given start vertex and print the visit order. Handle disconnected graphs. - Implement BFS to traverse a graph and print vertices level by level. - Find all connected components in an undirected graph using DFS/BFS.		

6. Perform topological sort on a directed acyclic graph (DAG) using Kahn's algorithm (BFS-based).
7. Implement DFS-based topological sort for a DAG.
8. Determine if a graph is bipartite (can be colored with 2 colors such that no adjacent nodes have the same color).
9. Given an $m \times n$ 2D binary grid which represents a map of '1's (land) and '0's (water), return the number of islands. An island is a group of adjacent '1's connected horizontally or vertically (not diagonally). You may assume all four edges of the grid are surrounded by water.

Unit 5	Advanced Data Structure	15 hours
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Advanced Data Structure: Binomial Heap, Operations on Binomial Heap, Trie data structures, Priority Queue, Disjoint data structure.

Problem solving:

1. Implement a binomial heap and write a function to merge two binomial heaps.
2. Extract the minimum key from a binomial heap and rebalance the heap.
3. Given an undirected graph, detect if it contains a cycle using Union-Find.
4. Design a Trie (prefix tree) data structure that supports the following operations:
 - insert(word): Inserts the string word into the trie.
 - search(word): Returns True if the string word is in the trie (i.e., it was inserted before), and False otherwise.
5. Design a program that simulates a job scheduler using a Priority Queue, where each job has a priority and a name. It should support three operations: (1) Add a Job with a given priority and name, (2) Execute the highest priority job, resolving ties by insertion order, and (3) Display all pending jobs in descending priority order, maintaining insertion order for equal priorities. The implementation must use an efficient priority queue structure like a heap with a tie-breaker mechanism for insertion order.

Total Lecture Hours	75 hours
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Text Book:

1. Cormen, Leiserson, Rivest, and Stein, Introduction to Algorithms (3 ed.), MIT Press, 2009. ISBN 978-0262033848.
2. Horowitz and Sahani, "Fundamentals of Data Structures", Galgotia Publications Pvt Ltd Delhi India.
3. Lipschutz, "Data Structures" Schaum's Outline Series, Tata McGraw-hill Education (India) Pvt. Ltd.
4. Reema Thareja, "Data Structure Using C" Oxford Higher Education.
5. AK Sharma, "Data Structure Using C", Pearson Education India.

Reference Book:

1. V. Aho, J. E. Hopcroft, and J. D. Ullman, Data Structures and Algorithms (1 ed.), Pearson, 1983.
2. Dasgupta, Papadimitrou and Vazirani, Algorithms (3 ed.), McGraw-Hill Education, 2006.
3. Horowitz, Sahni, and Rajasekaran, Computer Algorithms (2 ed.), Silicon Press, 2007.
4. Kleinberg and Tardos, Algorithm Design (1 ed.), Pearson, 2005.

Mode of Evaluation

MSE		CA			ESE	Total	
MSE1	MSE2	CA1	CA2	CA3 (ATTN)			
40	40	8	8	4			
80		20					

Course Code: IT107B	Course Name: Artificial Intelligence Essentials				L	T	P	C
					3	0	2	4
Pre-requisite: Python and Data Structure								
Course Objectives:								
To equip students with a foundational understanding of core AI concepts and hands-on proficiency in AWS AI services for designing, implementing, and evaluating intelligent solutions while preparing for the AWS Certified AI Practitioner exam.								
Course Outcome: After completion of the course, the student will be able to								
1. Understand AI fundamentals, its history, key concepts and ethical considerations. 2. Apply state-space search methods, expert-system paradigms, and Amazon Kendra for intelligent search solutions. 3. Use AWS Comprehend, Transcribe, and Textract to perform NLP tasks. 4. Design and implement generative AI applications using Amazon Bedrock and Amazon Q. 5. Design and deploy end-to-end AI solutions by integrating AWS services and formulate a strategy for AWS Certified AI Practitioner exam preparation.								



CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)													
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	
CO1	3	-	-	-	2	-	-	3	-	-	-	2	
CO2	3	3	2	2	3	-	-	-	-	-	-	2	
CO3	2	2	2	2	3	-	-	-	-	-	-	2	
CO4	2	2	3	2	3	-	-	-	2	-	2	2	
CO5	2	2	3	2	3	-	-	-	3	-	3	2	
Unit 1			Introduction to AI and AWS AI Services										15 hours
Introduction to AI: Evolution, AI from Turing Test to Humanoids, Applications, AI vs ML vs DL vs Data Science, Overview of AI Technologies: NLP, Computer Vision, Robotics, Future of AI: Tools, Trends, Ethics & Bias in AI, Introduction to AWS AI Practitioner Path, Getting Started with AWS cloud concepts. <i>Case Study: How Amazon Alexa uses AI.</i>													
Unit 2			AI Search Techniques and Rule-Based Systems										15 hours
Problem Solving: State Space, Search, Breadth-First Search (BFS), Depth-First Search (DFS), Heuristic Search: A*, Greedy Algorithm, Introduction to Rule-Based Expert Systems, Mapping AWS: Amazon Kendra for Intelligent Search. Getting started with Amazon Lex. <i>Case Study: Building a chatbot with AWS Lex.</i>													
Unit 3			Natural Language Processing (NLP) and AWS Tools										15 hours
Introduction to NLP: Tokenization, Stemming, Lemmatization, Sentiment Analysis, Using AWS Comprehend: Custom Classification and Entity Recognition, Introduction to Amazon Transcribe (Speech to Text), Amazon Textract: Process Documents with Synchronous and Asynchronous Operations. <i>Case Study: Automating customer feedback analysis using AWS Comprehend.</i>													
Unit 4			Introduction to Generative AI										15 hours
Generative AI: Basics, Applications, Foundation Models and Introduction to Amazon Bedrock, Planning a Generative AI Project, Building a Generative AI-Ready Organization, Amazon Q: Quick Introduction, Developer and Business. <i>Case Study: AI-generated product descriptions using Amazon Bedrock.</i>													
Unit 5			Real World AI Applications and Exam Preparation										15 hours
Generative AI Essentials on AWS, Business Intelligence with AWS QuickSight, Smart Applications: Recommendation Systems, Chatbots, Exam Prep Strategies for AWS AI Practitioner Certification, Practice Exam Questions. <i>Case Study: Personalized shopping experience using Generative AI.</i>													
Total Lecture Hours												75 hours	
Textbook:													
1. Russell, S., & Norvig, P. (2020). Artificial Intelligence: A Modern Approach (4th ed.). Pearson.													
2. Manish Kumar (2021). Artificial Intelligence: Principles and Applications. McGraw Hill Education.													
3. Vohra, D. (2022). Learning Amazon QuickSight. Packt Publishing.													
4. Tutorials Dojo. (2024). AWS Certified AI Practitioner Study Guide AIF-C01.													
5. Amazon Web Services. (2024). AWS Certified AI Practitioner (AIF-C01) Official Study Guide.													
References:													
1. https://portal.tutorialsdojo.com/product/study-guide-ebook-aws-certified-ai-practitioner-aif-c01/													
2. https://aws.amazon.com/certification/certified-ai-practitioner/													
3. https://docs.aws.amazon.com/													
Mode of Evaluation													
MSE		CA			ESE		Total						
MSE1	MSE2	CA1	CA2	CA3 (ATT)									
40	40	8	8	4									
		20			100		200						

Course Code: CS401L	Course Name: Design & Analysis of Algorithms	L	T	P	C
		3	0	0	3
Pre-requisite: The course requires a strong foundation in mathematics, algorithmic problem-solving, and proficient programming skills.					



Course Objectives:													
1. Develop a strong understanding of algorithm design techniques for efficient problem-solving.													
2. Analyze algorithm complexity and explore NP-completeness, approximation, and randomized algorithms.													
Course Outcome: After completion of the course, the student will be able to													
1. Analyse algorithm efficiency using recurrence relations and sorting techniques.													
2. Solve optimization problems with Greedy algorithms (MST, Shortest Path).													
3. Implement Dynamic Programming for Knapsack, Matrix Chain Multiplication, etc.													
4. Apply Backtracking and Branch & Bound to combinatorial problems.													
5. Understand NP-Completeness, Approximation, and Randomized algorithms.													
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)													
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	
CO1	3	3	2	2	2	-	-	1	-	-	1	2	
CO2	3	3	3	2	2	-	-	1	-	-	1	2	
CO3	3	3	3	3	2	-	-	1	-	-	1	2	
CO4	3	3	2	3	2	-	-	1	-	-	1	2	
CO5	3	3	2	3	2	-	-	1	-	-	1	2	
Unit 1		Introduction & Sorting in Linear Time											09 hours
Introduction: Algorithms, Algorithm Analysis, Recurrence Relations- substitution, Master’s method, Change of Variables, shell sort, Tim sort, Linear time sorting – Counting sort, Radix sort, Bucket sort.													
Unit 2		Greedy											09 hours
Greedy Algorithm: General Characteristics, Problem solving using Greedy Algorithm- Fractional Knapsack problem, Activity selection problem. Task scheduling with deadline and penalty problem, Minimum Spanning trees (Kruskal’s algorithm, Prim’s algorithm). Single source shortest paths - Dijkstra’s and Bellman Ford algorithms.													
Unit 3		Dynamic Programming											09 hours
Dynamic Programming: Introduction, The Principle of Optimality, Problem Solving using Dynamic Programming- optimal binary search trees, 0-1 Knapsack, Coin Change Problem, Assembly Line Scheduling, Longest Common Subsequence, Word break problem, Matrix chain multiplication, Rod cutting problem, All pair shortest paths – Warshal’s and Floyd’s algorithms, Resource allocation problem.													
Unit 4		Backtracking & Branch and Bound											09 hours
Back Tracking: Introduction, Subset sum problem, Permutation, N-queen problem, Graph Coloring, Hamiltonian Cycles and Sum of subsets, analysis of these problems. Branch and Bound: Traveling salesman problem.													
Unit 5		String Matching, Approximation Algorithm											09 hours
String Matching: Introduction, Naive string matching algorithm, Rabin-Karp algorithm, String Matching with finite automata, KMP (Knuth Morris Pratt) matching algorithm, Boyer Moore String matching. Approximation algorithms: Travelling Salesman problem, Hamiltonian problem, Vertex Cover Problem. Introduction to NP-Completeness: The class P and NP, Polynomial reduction, NP Completeness Problem, NP-Hard Problems. Randomized Algorithm.													
Total Lecture Hours											45 hours		
Textbook:													
1. Cormen, Leiserson, Rivest, and Stein, Introduction to Algorithms (3 ed.), MIT Press, 2009. ISBN 978-0262033848.													
Reference Books:													
1. Dasgupta, Papadimitrou and Vazirani, Algorithms (3 ed.), McGraw-Hill Education, 2006. ISBN 978-0073523408.													
2. Design and Analysis of Computer Algorithms by Aho, Hopcroft and Ullman, Pearson.													
3. Horowitz, Sahni, and Rajasekaran, Computer Algorithms (2 ed.), Silicon Press, 2007. ISBN 978-0929306414.													
4. Kleinberg and Tardos, Algorithm Design (1 ed.), Pearson, 2005. ISBN 978-0321295354.													
Mode of Evaluation													
MSE		CA			ESE		Total						
MSE1	MSE2	CA1	CA2	CA4 (ATT)									
30	30	6	6	3									
60		15			75		150						

Course Code: IT302L			Course Name: Computer Networks					L	T	P	C	
								3	0	0	3	
Pre-requisite: Computer Fundamentals												
Course Objectives:												
The objective of this course is to provide insight about computer network concepts and to gain comprehensive knowledge of the layered communication architectures, its functionalities, key protocols as well as security issues related with each layer.												
Course Outcome: After completion of the course, the student will be able to												
1. Apply the knowledge of networking concepts and functionality of physical layer. 2. Explore the concept of elementary data link layer protocol to build a robust network. 3. Analyze the concept of routing and IP addressing in network layer. 4. Examine the usage and working of transport layer. 5. Determine the performance of different protocols used at application layer.												
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)												
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	-	-	-	-	-	-	-	2
CO2	3	3	-	-	2	-	-	-	-	-	-	2
CO3	3	3	-	-	2	-	-	-	-	-	2	2
CO4	3	3	-	-	-	-	-	-	-	-	-	2
CO5	3	3	-	-	-	-	-	-	-	-	-	2
Unit 1	Introduction -Physical Layer										09 hours	
Introduction to computer networks: Network applications, Mode of communications, LAN, MAN, WAN, Internet, network hardware, network software, Design issues of layers, reference models: OSI, TCP/IP layers, and characteristics. Physical Layer: Network devices, Network topology, Transmission media, Signal transmission, Network performance and transmission impairments, the public switched telephone network, Switching techniques.												
Unit 2	Data Link Layer										09 hours	
Data Link layer: Design issues, Framing, Error Detection and Correction, Flow control (Elementary Data Link Protocols, Sliding Window protocols). Medium Access Control and Local Area Networks: Multiple access protocols: Random Access (ALOHA, CSMA, CSMA/CD, CSMA/CA), Controlled Access (Polling, Reservation, Token Passing), Channelization (FDMA, TDMA, CDMA). IEEE standards Wired LANs (Ethernet): Traditional Ethernet, Fast Ethernet, Gigabit Ethernet, Wireless LANs: IEEE 802.11 and Bluetooth.												
Unit 3	Network Layer										09 hours	
Network Layer: Network layer design issues, Logical addressing: Basic internetworking, Subnetting, Supernetting, NAT, network layer protocols, IPv4, IPv6 Routing: Static and dynamic routing, Routing algorithms and protocols (Shortest Path Algorithms, Flooding, Distance Vector Routing, Link State Routing, Hierarchical Routing, Broadcast Routing, Multicast Routing, Anycast Routing: RIP, OSPF, BGP)												
Unit 4	Transport Layer										09 hours	
Transport Layer : Process-to-process delivery, Transport layer protocols (UDP and TCP), SCTP, Connection management, Congestion control in TCP: Congestion Control: Open Loop, Closed Loop choke packets; Quality of service: techniques to improve QoS: Leaky bucket algorithm, Token bucket algorithm												
Unit 5	Application Layer										09 hours	
Session Layer: Roles and responsibility of session layer protocols Presentation Layer: Cryptography Symmetric-key cryptography, Asymmetric-key cryptography, Digital Signature, Compression: Lossless and Lossy Compression Application Layer: Domain Name System, World Wide Web and Hyper Text Transfer Protocol, Electronic mail, File Transfer Protocol, Simple Network Management Protocol, Telnet.												
Total Lecture Hours										45 hours		
Textbook:												
1. Behrouz Forouzan, “Data Communication and Networking” Fourth Edition-2006, Tata McGraw Hill 2. Andrew Tanenbaum “Computer Networks”, Fifth Edition-2011, Prentice Hall. 3. William Stallings, “Data and Computer Communication”, Eighth Edition-2008, Pearson.												
Reference Books:												
1. Kurose and Ross, “Computer Networking- A Top-Down Approach”, Eighth Edition-2021, Pearson.												

2. Peterson and Davie, "Computer Networks: A Systems Approach", Fourth Edition-1996, Morgan Kaufmann						
3. Behrouz Forouzan, "TCP/IP Protocol Suite", McGraw Hill.						
Mode of Evaluation						
MSE		CA			ESE	Total
MSE1	MSE2	CA1	CA2	CA (ATT)		
30	30	6	6	3		
60		15			75	150

Course Code: CS208B				Course Name: Web Technology				L	T	P	C		
								2	0	2	3		
Pre-requisite: Web Development, HTML, CSS, JavaScript													
Course Objectives:													
1. To familiarize students with the modern technologies used in web technologies.													
2. To realize the fundamentals of flutter & python web development frameworks.													
Course Outcome: After completion of the course, the student will be able to													
1. To understand advanced ES6 JavaScript features to develop modular and asynchronous applications.													
2. To build cross-platform mobile applications by applying Dart and Flutter with OOP and async concepts.													
3. To design and implement interactive Flutter applications using state management, navigation, and basic animations.													
4. To build web applications by applying Flask and Django, utilizing MVC architecture, form handling, and REST API development.													
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)													
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	
CO1	3	2	2	-	2	-	-	-	-	-	-	2	
CO2	3	2	3	-	2	-	-	-	2	-	-	2	
CO3	2	2	3	-	2	-	-	-	2	1	-	2	
CO4	3	3	2	2	2	-	-	-	2	1	-	2	
Unit 1			Advanced ES6 JavaScript										15 hours
Overview of ES6 features, Variable declarations: var vs let vs const, Arrow functions: syntax, single-line and multi-line arrow functions, lexical this keyword, Template literals, Array destructuring and object destructuring, default values, Rest operator (...) and spread operator with arrays and objects, Introduction to asynchronous programming, Promises: creating and consuming promises, then(), catch(), and basic error handling.													
Sample Problems:													
1. Write ES6 code Script to:													
a. Create a function using the rest operator to accept multiple numbers and return their sum.													
b. Merge two arrays using the spread operator.													
c. Copy and update an object using the spread operator.													
d. Demonstrate passing array elements as function arguments using spread.													
2. Write ES6 Promise that:													
e. Resolves after 2 seconds with a success message.													
f. Rejects if a condition fails.													
g. Consume the Promise using then() and catch().													
h. Display appropriate success or error messages.													
Unit 2			Fundamentals of Flutter and Dart Programming										15 hours

InSetup & Introduction: Introduction to Cross-Platform Development and flutter framework, Installing Flutter SDK, Android Studio Setup, basic arithmetic operations (+, -, *, /) using functions and switch statements.
 Dart Basics: Variables, Data Types, Control Flow (if/else, loops), Functions, Scope.
 OOP & Async: Classes, Objects, Inheritance, Mixins, Future, async/await.
 Widgets & Layouts: Stateless vs. Stateful Widgets, Container, Row, ListView, Image, Assets & Styling.

Sample Problems:

1. Building "Hello User" Console App, building a Simple Calculator: Dart command-line app implementing basic arithmetic operations (+, -, *, /) using functions and switch statements.
2. Basic Todo List (Logic Only): A command-line program that manages a list of tasks (add, remove, view) using Dart lists and loops.
3. Building OOP University Model: Create classes for Student, Course, and Professor using inheritance and encapsulation principles. Build a basic Dart script that uses async/await to simulate fetching data from a web source after a delay.

Unit 3	Input, Navigation UI and Animations	15 hours
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Interactivity & Input: Handling Input Forms, Handling Taps and Events
 State Management: Understanding Ephemeral State, setState()
 Navigation: push/pop, named routes, Common UI patterns: BottomNavBar, TabBar, Drawer. Lists & Grids: ListView, GridView, cards, Basic animations: AnimatedContainer, Hero. Introduction to themes, fonts and colors.
 Introduction to Android App Deployment and Publishing on Google Play: App packaging, signing, store listing, policies, and post-release maintenance.

Sample Problems:

1. Build a Profile Card/Business Card App: A purely static UI app using Column, Row, CircleAvatar, Icon, and Text widgets to design a visually appealing digital business card.
2. Build a BMI Calculator (UI Focus): A full UI with text fields for height/weight and a button. Use simple calculations and setState to display the result on the same screen.
3. Build a Basic Quiz App: A UI-focused app where users tap buttons to answer questions. Use setState to change the displayed question and options dynamically.

Unit 4	Web development with Python : Django & Flask	15 hours
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Introduction to Flask: Installation, basic routing, request and response handling, Jinja2 templating, form handling, validation, Building REST APIs using Flask.
 Introduction to Django: MVC architecture, installation, project setup, connecting to databases, Creating models, migrations, querying the databases, Django Views & Templates, URL routing, template rendering, form handling.

Sample problems:

1. Develop a Flask web application that demonstrates basic routing, request and response handling, and Jinja2 templating. The application should include a form with validation and expose a simple REST API that returns data in JSON format.
2. Develop a Django web application to demonstrate MVC architecture, project setup, URL routing, views, and template rendering. The application should include a form to accept user input and display the submitted data on a response page.

Total Lecture Hours	30 hours
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Textbook:

1. JavaScript: The Definitive Guide, David Flanagan
2. Flutter and Dart: Up and Running, Roopal Khurana and Dr. Deepti Chopra
3. Flutter for Beginners by Alessandro Biessek
4. Flask Web Development: Developing Web Applications with Python, Miguel Grinberg

Reference Books:

1. Flutter and Dart Cookbook: Developing Full-Stack Applications for the Cloud by Richard Rose
2. Mastering Dart & Flutter 2025

Mode of Evaluation

MSE		CA			ESE	Total
MSE1	MSE2	CA1	CA2	CA3 (ATTN)		
30	30	6	6	3		
60		15			75	150

Course Code: HS112L		Course Name: Universal Human Values						L	T	P	C	
								3	0	0	3	
Pre-requisite: NA												
Course Objectives:												
1. To help students distinguish between values and skills, and understand the need, basic guidelines, content, and process of value education.												
2. To help students initiate a process of dialog within themselves to know what they really want to be in their life and profession												
3. To help students understand the meaning of happiness and prosperity for a human being.												
4. To facilitate the students to understand harmony at all the levels of human living, and live accordingly.												
5. To facilitate the students in applying the understanding of harmony in existence in their profession and lead an ethical life.												
Course Outcome: After completion of the course, the student will be able to												
1. Understand the significance of value inputs in a classroom, distinguish between values and skills, understand the need, basic guidelines, content, and process of value education, explore the meaning of happiness and prosperity, and do a correct appraisal of the current scenario in the society.												
2. Distinguish between the Self and the Body, and understand the meaning of Harmony in the Self and the Co-existence of Self and Body.												
3. Understand the value of harmonious relationships based on trust, respect, and other naturally acceptable feelings in human-human relationships and explore their role in ensuring a harmonious society.												
4. Understand the harmony in nature and existence, and workout their mutually fulfilling participation in nature.												
5. Distinguish between ethical and unethical practices, and start working out the strategy to actualize a harmonious environment wherever they work.												
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)												
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	2	2	2	1	-	1	1
CO2	-	-	-	-	-	2	2	2	1	-	1	1
CO3	-	-	-	-	-	2	2	2	1	-	1	1
CO4	-	-	-	-	-	2	2	2	1	-	1	1
CO5	-	-	-	-	-	2	2	3	1	-	1	1
Unit 1	Introduction to Value Education										10 hours	
Understanding the need, basic guidelines, content, and process for Value Education, Self- Exploration-what is it? - its content and process; 'Natural Acceptance' and Experiential Validation -as the mechanism for self-exploration, Continuous Happiness, and Prosperity-A look at basic Human Aspirations, Right understanding, Relationship, and Physical Facilities-the basic requirements for fulfillment of aspirations of every human being with their correct priority, Understanding Happiness and Prosperity correctly – A critical appraisal of the current scenario, Method to fulfill the above human aspirations: understanding and living in harmony at various levels.												
Unit 2	Understanding Harmony in the Human Being										10 hours	
Understanding human being as a co-existence of the sentient 'I' and the material 'Body', Understanding the needs of Self ('I') and 'Body' - Sukh and Suvidha, Understanding the Body as an instrument of 'I' (I being the doer, seer, and enjoyer), Understanding the characteristics and activities of 'I' and harmony in 'I', Understanding the harmony of I with the Body: Sanyam and Swasthya; correct appraisal of Physical needs, the meaning of Prosperity in detail, Programs to ensure Sanyam and Swasthya.												
Unit 3	Understanding Harmony in the Family and Society										10 hours	
Harmony in Human-Human Relationship Understanding harmony in the Family-the basic unit of human interaction, Understanding values in the human-human relationship; meaning of Nyaya and program for its fulfillment to ensure Ubhay-tripti; Trust (Vishwas) and Respect(Samman) as the foundational values of relationship, Understanding the meaning of Vishwas; Difference between intention and competence, Understanding the meaning of Samman, Difference between respect and differentiation; the other salient values in a relationship, Understanding the harmony in the society (society being an extension of the family): Samadhan, Samridhi, Abhay, Sah- astitva as comprehensive Human Goals, Visualizing a universal harmonious order in society- Undivided Society (Akhand Samaj), Universal Order (Sarvabhaum Vyawastha) – from family to world family!												
Unit 4	Understanding Harmony in Nature and Existence										09 hours	

Whole existence as Co-existence Understanding the harmony in Nature, Inter connectedness, and mutual fulfillment among the four orders of nature- recyclability and self-regulation in nature, Understanding Existence as Co-existence (Sah-astitva) of mutually interacting units in all- pervasive space, Holistic perception of harmony at all levels of existence.		
Unit 5	Implications of the above Holistic Understanding of Harmony on Professional Ethics	06 hours
Natural acceptance of human values, Definiteness of Ethical Human Conduct, Basis for Humanistic Education, Humanistic Constitution and Humanistic Universal Order, Competence in Professional Ethics.		
Total Lecture Hours		45 hours
Textbook:		
1. R R Gaur, R Asthana, G P Bagaria, 2019 (2nd Revised Edition), A Foundation Course in Human Values and Professional Ethics. ISBN 978-93-87034-47-1, Excel Books, New Delhi.		
Reference Books:		
1. Ivan Illich, Energy & Equity, The Trinity Press, Worcester and Harper Collins, USA,1974.		
2. E.F. Schumacher, Small is Beautiful: a study of economics as if people mattered, Blond & Briggs, Britain,1973.		
3. A Nagraj, Jeevan Vidya EkParichay, Divya Path Sansthan, Amarkantak 1998.		
4. P L Dhar, RR Gaur, Science and Humanism, Commonwealth Publishers 1990.		
Mode of Evaluation		
MSE	CA	ESE
MSE1	CA1	Total
30	6	
30	6	
60	15	75
		150

Course Code: HS113L	Course Name: Aptitude-2							L	T	P	C	
								1	0	0	1	
Pre-requisite: NA												
Course Objectives:												
1. To provide adequate exposure to the students regarding the use of aptitude tests in the recruitment process and competitive examinations.												
2. To improve the logical & numerical ability of the students.												
Course Outcome: After completion of the course, the student will be able to												
1. Illustrate their comprehension by solving the given problems												
2. Apply the learned concepts to new problems and solve them aptly.												
3. Make use of their thought process to interpret and draw inferences from the given data to reach logical conclusions.												
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)												
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	-	1	-	1	-	-	-	-	-	1
CO2	1	1	-	1	-	2	-	-	-	-	-	1
CO3	1	1	-	1	-	1	-	-	-	-	-	2
Unit 1	Analytical Reasoning & Logical Puzzles										04 hours	
Definition and Introduction of Concept and Relation of Cube and Cuboids, Cut the cube in different layer and then solve questions accordingly. Problems related with open and closed dice.												
Unit 2	Syllogism										03 hours	
Understanding of Venn diagram, Problems related with Venn diagram, Statement and Conclusion, Syllogism and reverse syllogism.												
Unit 3	Clock and Calendar										04 hours	
Definition and Introduction of Concept and Relation of angle and time, Overtaking, overlapping, right-angle and straight Angle with respect to time, Error in clock (faster and slower), Correct time of clock, Mirror and Water Image of clock, Introduction of Calendar, Concept of Normal and Leap Year, Finding Odd days, Finding the day of the week of given date with and without reference.												
Unit 4	Data Interpretation and Critical Reasoning										04 hours	
Tables (Understand of Table, Fillers in table), Line Graph (Understand the graph, Percentage change, Ratio based comparison), Bar Graph (Type of Bar Graph, Average and Comparison, Stacked Bar Graph), Pi Chart (Conversion of Percentage and Degree, Fillers in Pie chart, Multiple Pie chart), Mixed Graph (problems related with combination of various charts) Critical												

Reasoning: Assumptions, Cause and Effect, Assertion and Reason, Statement and Inference					Total Lecture Hours	15 hours
Useful resources:						
1. “A Modern Approach to Verbal & Non-Verbal Reasoning” by R.S. Aggarwal, S. Chand Publication.						
2. https://www.geeksforgeeks.org/most-important-aptitude-topics-for-placements/						
Reference Books:						
1. "How to Prepare for Logical Reasoning for the CAT" by Arun Sharma, TMH Publication.						
2. https://www.indiabix.com/logical-reasoning/questions-and-answers/						
3. https://testbook.com/placement-aptitude/test-series						
Mode of Evaluation						
MSE		CA			ESE	Total
MSE1	MSE2	CA1	CA2	CA3 (ATT)		
-	-	10	10	5		
-		25			25	50

Course Code: HS114L				Course Name: Soft Skills Essentials-2				L	T	P	C	
								1	0	0	NC	
Pre-requisite:												
• Successful completion of the subject ‘Soft Skills Essentials-1’ of the third semester.												
Course Objectives:												
To strengthen students’ professional communication, cultural intelligence, and emotional awareness through advanced speaking activities, scenario-based discussions, and digital literacy tasks, equipping them for diverse workplace interactions.												
Course Outcome: After completion of the course, the student will be able to												
1. Apply advanced communication strategies that include vocabulary enhancement, storytelling to improve their cultural sensitivity (DEI).												
2. Demonstrate prompt writing for AI-based tools and create effective elevator pitches to convey ideas with clarity and impact.												
3. Exhibit interpersonal effectiveness by navigating negotiation, persuasion, and emotional intelligence in professional contexts												
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)												
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	1	3	-	2
CO2	-	-	-	-	-	-	-	-	1	3	-	2
CO3	-	-	-	-	-	1	-	-	1	3	-	2
Unit 1	Advanced Communication and Cultural Sensitivity										7 hours	
Vocabulary Enhancement through Gamification, Story Coining and Presentations Understanding Cross-Cultural, Communication (DEI) using Case Studies, Duo Lingo English Proficiency Tests												
Unit 2	Professional Expression and Digital Literacy										4 hours	
TMAY through Driver’s test, Writing Effective Prompts on Various LLMs, Duo Lingo English Proficiency Tests												
Unit 3	Interpersonal Effectiveness and Emotional Intelligence										4 hours	
Negotiation & Persuasion Role Plays, Developing Emotional Intelligence via Scenario-Based Discussions												
Total Lecture Hours										15 hours		
Useful Resources:												
1. https://youtu.be/5Wr-uaGzY7c												
2. https://youtu.be/NcCwlqBapHo												
3. https://youtu.be/SKNmQPIBPiI												
4. RAISEC - B. Tech. MCA - Introduction												
5. RAISEC - B. Tech. MCA - Social Personality Type												
6. RAISEC - B. Tech. MCA - Enterprising Personality Type												
7. RAISEC - B. Tech. MCA - Conventional Personality Type												
Mode of Evaluation												
MSE	CA				ESE		Total					



MSE1	MSE2	CA1	CA2	CA3		
-	-	10	10	5		
-		25			25	-

Course Code: IT315E				Course Name: Generative AI Foundations & Applications						L	T	P	C
										3	0	2	4
Pre-requisite: Python Programming, Basics of Statistics, Basic Mathematics, Basics of Machine Learning													
Course Objectives:													
To equip students with foundational and applied knowledge of Generative AI, NLP, and Large Language Models (LLMs), with hands-on experience in building GenAI solutions using Python frameworks such as Hugging Face, LangChain, and Lang Graph. Students will learn prompt engineering, fine-tuning, retrieval-augmented generation (RAG), and responsible AI deployment strategies.													
Course Outcome: After completion of the course, the student will be able to													
1. Understand core NLP concepts, word embeddings, and transformer-based architectures including encoder-decoder models.													
2. Explain the structure and functionality of foundational LLMs such as GPT, BERT, LLaMA and utilize inference parameters to control outputs.													
3. Apply prompt engineering and fine-tuning techniques (LoRA, PEFT) for task-specific adaptation of language models.													
4. Design GenAI applications using RAG architecture, vector databases, and multi-agent systems with LangChain and LangGraph.													
5. Analyze and mitigate LLM risks such as bias and hallucination using evaluation techniques and AI safety framework													
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)													
CO-PO Mapping		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		3	3	3	3	3	2	3	3	3	3	3	3
CO2		3	3	3	3	3	2	3	2	3	3	3	3
CO3		3	3	3	3	3	2	3	3	3	3	3	3
CO4		3	3	3	3	3	3	3	3	3	3	3	3
CO5		3	3	3	3	3	3	3	3	3	3	3	3
Unit 1		Introduction to NLP and Transformer Architectures										15 hours	
Introduction to NLP: Tokenization, Lemmatization, POS tagging, Stopwords removal Word Embeddings: One-hot, Word2Vec, GloVe, FastText Neural Networks in NLP: RNNs, LSTMs, GRUs Transformer Architecture: Attention Mechanism, Positional Encoding Encoder and Decoder models: Transformer blocks in NLP Basic NLP Use Cases: Text classification, Sentiment analysis, NER Implementation: Using Hugging Face Transformers, SpaCy, NLTK for pipeline construction Case Study: Building a Named Entity Recognition system using Hugging Face Transformers													
Unit 2		Generative AI and Foundation Models										15 hours	
Introduction to Generative AI: Concepts and Applications Multimodal Generative Models: Overview of CLIP, DALL·E Foundation Models Overview: BERT (encoder), GPT (decoder), LLaMA, OLMA Comparative Architecture Analysis: Encoder-only, Decoder-only, and Encoder-Decoder Inference Tuning Parameters: Temperature, Top-k, Top-p, Max tokens Hands-on Text Generation using OpenAI and Hugging Face APIs Case Study: Generating product reviews using GPT-2 and LLaMA Domain-specific response generator (e.g., medical, legal, HR),AI Study Companion for Engineering Students.													
Unit 3		Prompt Engineering and Fine-Tuning Techniques										15 hours	
Introduction to Prompt Engineering: Zero-shot, One-shot, Few-shot Advanced Techniques: Chain-of-Thought (CoT), ReAct prompting, System role promptingOutput schema & structured prompting Prompt Chaining and Templates using LangChain Fine-Tuning Overview: Full Model vs. Parameter Efficient (PEFT, LoRA, QLoRA)													

Fine-Tuning vs. Prompting: When to use which?										
Model Evaluation: Perplexity, BLEU, ROUGE metrics										
Implementation using Hugging Face PEFT and Transformers										
Case Study: Fine-tuning BERT for custom domain classification , College Exam Question Solver										
Unit 4		RAG Architecture and Multi-Agent Systems						15 hours		
RAG Introduction: Bridging retrieval and generation										
Embedding Models: OpenAI Embeddings, Sentence Transformers										
Vector Databases: FAISS, ChromaDB, Pinecone – Indexing and Querying										
LangChain for RAG: Document loading, Retriever chains, Prompt templates										
Introduction to Agents: Tool usage, reasoning and acting										
LangGraph and MCP Protocol: Coordinating multi-agent systems										
Building end-to-end RAG applications with LangChain and LangGraph										
Case Study: Building a document-based chatbot with RAG and FAISS										
Unit 5		LLM Limitations, Evaluation & Responsible AI						15 hours		
Understanding LLM Challenges: Bias, Hallucination, Overfitting										
Bias Mitigation: Dataset curation, response filtering, model instruction tuning										
Hallucination Mitigation: RAG pipelines, retrieval anchoring										
Evaluation Methods: Automated (BLEU, Perplexity), Human-centric (coherence, helpfulness)										
Ethical AI Principles: Fairness, Transparency, Accountability										
Guardrails and Safety Tools: Moderation APIs, Red teaming, Guardrails AI										
LLM Governance Frameworks and Open-source safety tooling										
Case Study: Analyzing and mitigating hallucinations in GPT-3 generated content.										
Total Lecture Hours								75 hours		
Textbook:										
1. Jurafsky & Martin, Deep Learning (Goodfellow), Hugging Face NLP										
2. Tunstall et al. (Transformers), Foster (GenAI), Rothman										
3. Hugging Face, PEFT Docs, Chip Huyen										
4. Tutorials Dojo. (2024). AWS Certified AI Practitioner Study Guide AIF-C01.										
5. Amazon Web Services. (2024). AWS Certified AI Practitioner (AIF-C01) Official Study Guide.										
Reference Books:										
1. Hugging Face Transformers										
2. OpenAI API Docs										
3. LangChain Docs										
4. LangGraph Docs										
5. Scikit-learns										
6. PEFT (Parameter-Efficient Fine-Tuning)										
7. FAISS (Facebook AI Similarity Search)										
Mode of Evaluation										
MSE		CA			ESE	Total				
MSE1	MSE2	CA1	CA2	CA3						
40	40	8	8	4						
80		20			100	200				

Course Code: CS321E	Course Name: Foundation of iOS App Development				L	T	P	C
					3	0	2	4
Pre-requisite: NA								
Course Objectives:								
1. To develop a strong foundation in Swift programming by understanding syntax, control flow, functions, collections, and object-oriented concepts required for iOS application development.								
2. To design and build basic iOS applications using UIKit by gaining hands-on experience with Xcode, view controllers, table views, collection views, and Auto Layout for adaptive user interfaces.								
Course Outcome: After completion of the course, the student will be able to								
1. To understand iOS ecosystem and Xcode tools								
2. To apply Swift programming fundamentals								

3. To implement control flow and functions in Swift 4. To develop UI using UIKit components 5. To build adaptive iOS apps using Auto Layout												
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)												
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	-	-	2	-	-	-	-	-	-	1
CO2	3	2	2	-	2	-	-	-	-	-	-	1
CO3	3	3	2	1	2	-	-	-	-	-	-	1
CO4	2	2	3	1	3	-	-	-	-	-	-	1
CO5	2	2	3	1	3	-	-	-	-	-	-	2
Content Covered: Introduction to iOS Ecosystem, Xcode Interface, Project Structure Swift Basics – Syntax, Variables, Data Types Control Flow – Conditions, Loops, Switch Functions, Closures (Basics), Error Handling Collections in Swift – Arrays, Dictionaries, Sets Object-Oriented Swift – Classes, Structs, Enums Optionals, Protocols, Extensions UIKit Essentials – Views, View Controllers, Storyboards TableView & CollectionView (Intro + Hands-on) Auto Layout Basics – Constraints, Adaptive UI Mini App Project (Calculator / Notes / Gallery)												
Total Lecture Hours											75 hours	
Textbook & Reference Books: 1. Develop in Swift (Official Apple Books) – Apple’s official series for learning Swift and iOS development (like Fundamentals and Data Collections). 2. Swift Apprentice (Kodeco) – Strong foundational book on Swift language and syntax (updated regularly).												
Mode of Evaluation												
MSE		CA			ESE	Total						
MSE1	MSE2	CA1	CA2	CA3								
40	40	8	8	4								
80		20			100	200						

Course Code: IT311E	Course Name: Cloud Computing & Operations								L	T	P	C
									3	0	2	4
Pre-requisite: AWS Cloud Foundation												
Course Objectives:												
The Primary objective of this course is to equip learners with the skills, tools, and operational mindset required to implement, monitor, automate, and scale cloud environments with a strong focus on AWS Cloud Operations (CloudOps).												
Course Outcome: After completion of the course, the student will be able to												
1. Design and implement cloud monitoring strategies using metrics, logs, alarms, and distributed tracing.												
2. Architect reliable and scalable cloud systems with disaster recovery and business continuity plans.												
3. Deploy and provision cloud resources using DevOps practices and Infrastructure-as-Code automation.												
4. Apply cloud security controls following compliance, policy, identity, and data-protection best practices.												
5. Build and troubleshoot cloud networking topologies with load balancing, routing, and content delivery.												
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)												
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	3	-	-	-	2	2	2	3
CO2	3	3	2	2	3	-	-	-	2	2	2	3
CO3	3	3	2	2	3	-	-	-	2	2	2	3
CO4	3	3	2	2	3	-	-	2	2	2	2	3
CO5	3	3	2	2	3	-	-		2	2	2	3

Unit 1	Monitoring, Logging, Analysis, Remediation & Performance Optimization					15 hours				
CloudOps fundamentals: cloud operational visibility, global infrastructure awareness, and telemetry-driven decision making. Monitoring: CloudWatch Metrics, KPI thresholds (CPU, latency, request count), and alarm-based incident triggers. Logging: CloudWatch Logs with Log Insights, retention policies, and centralized ingestion. Distributed tracing: AWS X-Ray service maps, latency hotspot detection, and dependency analysis. Remediation & optimization: auto-scaling, predictive monitoring, and Trusted Advisor recommendations for performance and cost. <i>Case Study: Application Performance Monitoring using CloudWatch Metrics, Logs, and Dashboards</i>										
Unit 2	Reliability & Business Continuity					15 hours				
Reliability engineering: multi-AZ deployments, active-active vs. active-passive architecture, and cross-region replication. Disaster recovery: RPO/RTO planning with pilot-light, warm standby, and hot failover strategies. Business continuity: snapshot automation, versioned backups, restore validation, and lifecycle retention. Operational robustness: health checks, rollback readiness, and controlled failure injection. <i>Case Study: Multi-AZ Database Failover using Amazon RDS Automated Backups</i>										
Unit 3	Deployment, Provisioning & Automation					15 hours				
Infrastructure-as-Code: CloudFormation/Terraform templates, environment parity, and drift detection. CI/CD deployment: rolling updates, blue-green staging, and canary release patterns with automated rollback. Provisioning pipelines: compute, container orchestration, and serverless rollout strategies. Secret handling: Parameter Store/Secrets Manager with role-based access. <i>Case Study: Continuous Deployment Pipeline using CodeCommit, CodeBuild, and CodePipeline</i>										
Unit 4	Security & Compliance					15 hours				
Access governance: IAM policies, role scoping (RBAC/ABAC), cross-account federation, and privilege boundaries. Data protection: KMS encryption at rest, TLS enforcement in transit, and secret rotation. Threat detection: GuardDuty anomaly signals, CloudTrail forensic logs, and Macie PII classification. Compliance posture: HIPAA/SOC/PCI alignment through audit logs, bucket policy hardening, and alert-driven remediation. <i>Case Study: Secure Storage Management using S3 Bucket Policies, IAM Roles, and Encryption</i>										
Unit 5	Networking, Content Delivery, Certification Prep & Capstone					15 hours				
Network architecture: VPC segmentation, routing tables, NAT/IGW, and perimeter defense using security groups/NACLs. Hybrid connectivity: VPN and DirectConnect with private endpoints for internal services. Traffic distribution: ALB/NLB/GLB load-balancing strategies and health-based routing. Content delivery: Route53 latency-aware resolution and CloudFront edge caching for global acceleration. CloudOps consolidation: monitoring integration, IaC automation, cost governance, identity boundaries, and network scaling patterns. Exam-style scenario analysis: fault tracing, DR validation, and architectural trade-off reasoning. <i>Case Study: Global Content Delivery Optimization using S3 + CloudFront Distribution</i> <i>Case Study: End-to-End Subscription Platform using IaC, CI/CD, and CloudWatch Observability</i>										
Total Lecture Hours						75 hours				
Textbook:										
1. Kavis, M. (2019). Architecting the Cloud: Design Decisions for Cloud Computing Service Models (SaaS, PaaS, and IaaS).										
2. Wittig, A. & Wittig, M. (2023). Amazon Web Services in Action (3rd ed.).										
Reference Books:										
1. Yevick, A., & Yevick, D. (2022). Terraform: Up & Running: Writing Infrastructure as Code (3rd ed.). O'Reilly Media.										
2. Beyer, B., Jones, C., Petoff, J., & Murphy, N. (2016). Site Reliability Engineering: How Google Runs Production Systems. O'Reilly Media.										
Mode of Evaluation										
MSE		CA			ESE	Total				
MSE1	MSE2	CA1	CA2	CA3						
40	40	8	8	4						
80		20			100	200				

Course Code: IT318E			Course Name: Cloud Systems Development						L	T	P	C
									3	0	2	4
Pre-requisite: AWS Cloud Foundation												
Course Objectives:												
The Primary objective of this course is to introduce cloud-native application development on AWS, focusing on building backend logic with managed services, storing data using cloud databases, and integrating applications through APIs and serverless components with a strong focus on development workflows, enabling learners to design, build scalable applications in the AWS environment.												
Course Outcome: After completion of the course, the student will be able to												
1. Develop cloud-native applications using AWS services. 2. Apply secure development practices with policies, authentication tokens, and access boundaries. 3. Manage state and persistence focusing on efficient querying, indexing, and data modeling. 4. Implement application-level logging, error handling, and tracing to improve reliability, debugging, and real-time insights. 5. Implement serverless and event-driven workflows, enabling functions to react to events, API calls, or data changes.												
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)												
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2	3	-	-	-	2	2	2	3
CO2	3	3	3	2	3	-	-	1	2	2	2	3
CO3	3	3	3	2	3	-	-	-	2	2	2	3
CO4	3	3	3	2	3	-	-	-	2	3	2	3
CO5	3	3	3	2	3	-	-	-	2	2	2	3
Unit 1	Cloud-Native Application Development with AWS Managed Services										25 hours	
Application logic built on managed services using AWS Lambda, API Gateway, DynamoDB, and S3. Focus on event-driven workflows, CRUD operations through service APIs, and AWS SDK integration for programmatic service access and data handling. Emphasis on reducing infrastructure overhead with serverless components, enabling concentration on business logic, modular service composition, and scalable cloud-native design patterns.												
Case Study: Serverless Task Manager using Lambda, API Gateway, and DynamoDB												
Unit 2	Secure Application Development with IAM, Policies, and Service Roles										15 hours	
Development with least-privilege IAM access, role-based permissions, and policy enforcement, using temporary credentials, API authentication, and encrypted storage. Emphasis on secure resource access from applications without exposing keys, avoiding public endpoints, and applying identity-driven security across services.												
Case Study: Secure File Upload App using IAM Roles, S3 Encryption, and Access Controls												
Unit 3	Data Persistence and State Management using AWS Storage and Databases										10 hours	
Structuring application data using DynamoDB, RDS, and S3. Focus on partition key design, secondary indexes, transactional operations, and simple schema modeling, while handling unstructured content storage. Emphasis on performance, durability, and efficient access patterns for scalable cloud-native applications.												
Case Study: Campus Notes Repository using DynamoDB + S3 for Structured and File-Based Storage												
Unit 4	Application Monitoring, Logging, and Error Diagnosis										10 hours	
Troubleshooting runtime issues using CloudWatch Metrics, function logs, and request tracing. Analyzing performance bottlenecks, input errors, and service latency. Integrating logs directly from Lambda or service APIs to refine application behavior.												
Case Study: API Error Debugging using CloudWatch Logs and AWS X-Ray												
Unit 5	Scalable Application Architecture with Modular and Event-Driven Components										15 hours	
Designing lightweight architectures that break functionality into independent services. Connecting components through event triggers, queues, or HTTP endpoints. Avoiding monolithic design and enabling incremental scaling as usage grows.												
Case Study: Simple Event Bulletin Platform using Lambda, DynamoDB Streams, and API Gateway												
Total Lecture Hours										75 hours		
Textbook & Reference Books:												
1. Various Authors (2024). <i>Serverless Development on AWS</i> . O'Reilly Media. 2. Wittig, M., & Wittig, A. (2021). <i>Amazon Web Services in Action (2nd ed.)</i> . Manning Publications. 3. Poccia, D. (2020). <i>AWS Lambda in Action: Event-Driven Serverless Applications</i> . Manning Publications. 4. Sbarski, P., Cui, Y., & Nair, A. (2022). <i>Serverless Architectures on AWS (2nd ed.)</i> . O'Reilly Media.												

Mode of Evaluation						
MSE		CA			ESE	Total
MSE1	MSE2	CA1	CA2	CA3		
40	40	8	8	4		
80		20			100	200

Course Code: AI103E			Course Name: Cloud Foundations					L	T	P	C	
								3	0	2	4	
Pre-requisite: Computer Fundamentals												
Course Objectives:												
This course introduces the fundamentals of cloud computing and AWS services, providing hands-on experience with compute, storage, networking, and security components. It aims to develop skills for monitoring and managing cloud environments, explore cloud adoption strategies, and provide students with AWS foundational and cloud career roles.												
Course Outcome: After completion of the course, the student will be able to												
1. Understand the fundamentals of cloud computing and core AWS services.												
2. Configure AWS compute, storage, networking, and database services.												
3. Implement security best practices using IAM, encryption, and network monitoring tools.												
4. Apply cloud adoption frameworks and infrastructure as code (IaC) strategies in cloud environments.												
6. Demonstrate proficiency in AWS learning tools for industry recognition.												
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)												
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	2	-	-	1	1	2	-	2
CO2	3	3	2	2	2	-	-	1	3	2	-	2
CO3	3	3	2	2	2	-	-	1	3	2	-	2
CO4	3	2	2	2	2	-	-	1	3	2	1	2
CO5	3	3	2	2	2	-	-	1	3	2	2	2
Unit 1	Introduction to Cloud Fundamentals and AWS Core Services										15 hours	
Introduction and Walkthrough of Skill Builder, Job Roles in the Cloud, Introduction to Amazon EC2: Basic overview of launching, resizing, managing, and monitoring an Amazon EC2 instance, Introduction to AWS Identity and Access Management (IAM): add users to groups, manage passwords, log in with IAM-created users, and see the effects of IAM policies on access to specific services. Case Study: Launching a Basic Web Server using EC2												
Unit 2	AWS Infrastructure, Storage, and Cost Management Essentials										15 hours	
AWS Networking Basics, getting started with AWS Storage, data transfer and data protection. Introduction to AWS Billing and Cost Management, AWS Database Offering, AWS Technical Essentials: fundamental AWS concepts related to compute, database, storage, networking, monitoring, and security. Configuring and deploying VPCs with multiple subnets. Case Study: Create a Secure Storage and Backup System using Amazon S3												
Unit 3	Advanced AWS Compute, Storage, and Cloud Adoption Strategies										15 hours	
Built with Amazon EC2: configure, launch, and securely access EC2 instances, AWS Lambda Foundations, Introduction to Cloud adoption framework(CAF), Amazon Elastic Block Store (EBS): relational and non-relational databases, enterprise applications, containerized applications, big data analytics engines, file systems, and media workflows., AWS Network – Monitoring and Troubleshooting. Case Study: Serverless File Processing System using AWS Lambda												
Unit 4	AWS Performance Optimization and Security Essentials										15 hours	
Introduction to Elastic Load Balancer, testing a Network Elastic Load Balancer, Introduction to CloudFront, Data Encryption in AWS, Server Side Encryption (SSE) for Amazon S3, AWS Key Management Service (KMS), and Amazon DynamoDB Encryption Client, AWS Encryption SDK to encrypt and decrypt data. Case Study: Deploy a Scalable Application with Security using the concepts of Elastic Load Balancer(ELB) and CloudFront.												
Unit 5	AWS Learning Tools and Infrastructure as Code (IaC) Foundations										15 hours	
Introduction to AWS CloudFormation, launch WordPress on Amazon Web Services, Cloud Essentials - Knowledge Badge, AWS services, security, architecture, pricing, and support, AWS Cloud Quest Case Study: Deploy a Full Stack Application using AWS CloudFormation Exam readiness: AWS Cloud Practitioner Certificate (CLF-C02)												
Total Lecture Hours										75 hours		

Textbook:

1. Piper, B., & Clinton, D. (2020). AWS Certified Solutions Architect Official Study Guide: Associate Exam. Wiley.
2. Wittig, M., & Wittig, A. (2018). Amazon Web Services in Action (2nd ed.). Manning Publications.
3. Wilkins, M. (2021). Learning Amazon Web Services (AWS): A Hands-On Guide to the Fundamentals of AWS Cloud. Addison-Wesley Professional.

Reference Books:

1. <https://awsacademy.instructure.com/courses/100702>
2. https://www.youtube.com/watch?v=UmQnenLf_Cs

Mode of Evaluation

MSE		CA			ESE	Total	
MSE1	MSE2	CA1	CA2	CA3			
40	40	8	8	4			
80		20			100	200	

3. Practical's Courses Detail Syllabus

Course Code: IT301P		Course Name: Database Systems Lab								L	T	P	C
										0	0	2	1
Pre-requisite: Concepts of any programming language													
Course Objectives:													
1. Develop Hands-on SQL Skills													
2. Design and Normalize Databases													
3. Work with PL/SQL and Advanced Database Concepts													
Course Outcome: After completion of the course, the student will be able to													
1. Design Logical and Conceptual database schema for real life problem using ERD													
2. Apply SQL to store, retrieve and manipulate data in relational database													
3. Apply PL/SQL to solve real world database management and automation task													
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)													
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	
CO1	2	2	3	1	3	1	1	1	2	2	1	2	
CO2	3	3	2	1	3	1	1	1	1	1	1	2	
CO3	3	3	2	1	3	1	1	1	1	2	1	2	
List Of Practical's (Indicative & Not Limited To)													
1. Creating Entity-Relationship Diagram using case tool (STAR UML)													
[Group formation and assigned project to submit a report at the end of the semester as project-based learning]													
1	Insurance Database												
2	Banking Enterprise Database												
3	Supplier Database												
4	Student Faculty Database												
5	Airline Flight Database												
6	Order Processing Database												
7	Book dealer Database												
8	Student Enrolment Database												
9	Movie Database												
10	College Database												

11	Hostel Database
12	Library Database
13	Clinic Database
14	Medical Store Database
15	Exam Process Database

2. Introduction to date types and database implementation using Create, insert, and Basic SQL SELECT statements. Create these two tables with following specifications and insert data in the table:

Table Name: Client master

Attribute	Data Type	Size
Client_no	number	10
Client_Name	Varchar2	20
City	Varchar2	15
State	Varchar2	15
Pin	Number	6
Balance_due	Number	10,2

Data for Client master:

CLIENT NO	Client NAME	CITY	STATE	PIN CODE	BAL DUE
0001	Ivan	Bombay	Maharastra	400057	15000
0002	Vandura	Madras	Tamilnadu	980001	0
0003	Pramod	Bombay	Maharastra	400057	5000
0004	Basu	Bombay	Maharastra	400056	0
0005	Ravi	Delhi	Null	100001	2000
0006	Rukmini	Bombay	Maharastra	900050	0

PRODUCT MASTER

COLUMN	DATA TYPE	Size
PRODUCT_NO	VARCHAR2	6
DESCRIPTION	VARCHAR2	20
PROFIT%	NUMBER	10
QTY_ON_HAND	NUMBER	10
ORDER_LEVEL	NUMBER	10
SELL_PRICE	NUMBER	10
COST_PRICE	NUMBER	10

Data for Product Master Table

Product No	Description	Profit %	Qty on hand	Reorder level	Sell price	Cost price
P00001	1.44 floppies	5	100	20	525	500
P03453	Monitors	6	10	3	12000	11200
P06734	Mouse	5	20	5	1050	500
P07865	1.22 floppies	5	100	20	525	500
P07868	Keyboards	2	10	3	3150	3050
P07885	CD drive	2.5	10	3	5250	5100
P07965	540 HDD	4	10	3	8400	8000
P07975	1.44 Drive	5	10	3	1050	1000
P08865	1.22 Drive	5	2	3	1050	1000

Perform following queries on the above data:

- Find out the name of all the clients.
- Retrieve the list of names and cities of all the clients.
- List all the clients who are located in Bombay.
- Display the information for client no 0001 and 0002.
- Find the list of all clients who stay in city 'Bombay' or 'Delhi' or 'Madras'.
- List the name, city, and state of clients not in state of 'Maharashtra'

3. To manipulate data in the existing tables**Using the table client master and product master answer the following queries:**

- Delete the record of Client no. 0001 from the Client master table.
- Change the city of Client no. 0005 to 'Bombay'.
- Change the balance due of Client no. 0002 to 1000.
- Find out the clients who stay in a city or state where second letter is a.
- Calculate the average balance due of all the clients.
- Change the selling price of 1.44 floppy drive to Rs. 1150.00.
- Count the number of products having price greater than or equal to 1500.

4. To create, manage tables with constraints and alter the structure of tables.**CREATE THE TABLES WITH FOLLOWING SPECIFICATIONS AND CONSTRAINTS:****TABLE NAME: SALES MASTER**

ATTRIBUTE	DATA TYPE	SIZE	CONSTRAINT
SALESMAN NO	VARCHAR2	6	PRIMARY KEY, FIRST LETTER IS 'S'
SALES NAME	VARCHAR2	20	NOT NULL
ADDRESS	VARCHAR2	20	NOT NULL
CITY	VARCHAR2	20	---
STATE	VARCHAR2	20	---
PINCODE	NUMBER	6	---
SAL AMT	NUMBER	8,2	NOT NULL, CAN'T BE ZERO
Tgt to get	NUMBER	6,2	NOT NULL, CAN'T BE ZERO
Ytd sales	NUMBER	6,2	NOT NULL, CAN'T BE ZERO
Remark	VARCHAR2	30	

TABLE NAME: SALES ORDER

ATTRIBUTE	DATA TYPE	SIZE	CONSTRAINT
S ORDER No	VARCHAR2	6	PRIMARY KEY, FIRST LETTER IS 'O'
S ORDER DATE	DATE	---	---
CLIENT NO	NUMBER	10	FOREIGN KEY FROM CLIENT MASTER
SALESMAN NO	VARCHAR2	26	FOREIGN KEY FROM SALES MASTER
DELIVERY TYPE	CHAR	1	P FOR PARTIAL AND F FOR FULL, DEFAULT IS F
BILLED YN	CHAR	1	'Y' FOR YES AND 'N' FOR NO
DELIVERY DATE	DATE	---	CAN'T BE LESS THAN S ORDER DATE
ORDER STATUS	VARCHAR2	10	IN(IN-PROCESS, FULFILLED, BACK ORDER, CANCELLED)

TABLE NAME: Sales_order_detail

Column	Datatype	Size	Attributes
S_order_no	Varchar2	6	PK/FK references s_order_no of sales_order
Product_no	Varchar2	6	PK/FK references product_no of product_master
Qty_order	Number	8	
Qty_disp	Number	8	
Product_rate	Number	10,2	

DATA OF SALES MASTER:

Sales No.	Sales_Name	Address	City	Pincode	State	Salamt	Tgt_to_get	Ytd_sales	Remark
-----------	------------	---------	------	---------	-------	--------	------------	-----------	--------

S00001	Kiran	A/14 worli	Bombay	400002	MAH	3000	100	50	Good
S00002	Manish	65, Nariman	Bombay	400001	MAH	3000	200	100	Good
S00003	Ravi	P-7, Bandra	Bombay	400032	MAH	3000	200	100	Good
S00004	Ashish	A/5 Juhu	Bombay	400044	MAH	3500	200	150	Good

DATA OF SALES ORDER

S_order_no	S_order_date	Client no	Dely type	Bill yn	Salesman no	Delay date	Orderstatus
O19001	12-Jan-96	1	F	N	50001	20-Jan-96	IP
O19002	25-Jan-96	2	P	N	50002	27-Jan-96	C
O16865	18-Feb-96	3	F	Y	500003	20-Feb-96	F
O19003	03-Apr-96	1	F	Y	500001	07-Apr-96	F
O46866	20-May-96	4	P	N	500002	22-May-96	C
O10008	24-May-96	5	F	N	500004	26-May-96	IP

Data for sale order detail

S_order_no	Product no	Qty_order	Qty_disp	Product_rate
O19001	P00001	4	4	525
O19001	P07965	2	1	8400
O19001	P07885	2	1	5250
O19002	P00001	10	0	525
O46865	P07868	3	3	3150
O46865	P07885	10	10	5250
O19003	P00001	4	4	1050
O19003	P03453	2	2	1050
O46866	P06734	1	1	12000
O46866	P07965	1	0	8400
O10008	P07975	1	0	1050
O10008	P00001	10	5	525

1. Make client_no primary key in client_master.
 2. Add new column phone_number in client_master table.
 3. Add not null constraint in product master with columns : description, profit_percent, sellprice, costprice
 4. Change size of client_no field in client_master.
 5. Add check constraint to product_master such that sellprice is always greater than costprice.
 6. Select produc_no,description where profit percent is between 20 and 30 both inclusive.
5. To implement join concepts.
 1. Find out the product which has been sold to 'Ivan'.
 2. Find out the product and their quantities that will have to be delivered.
 3. Find out the names of clients who have purchased 'CD DRIVE'
 4. List the product_no and s_order_no of customers having qty ordered less than 5 from the order details table for the product '1.44 floppies'.
 5. Find the product and their quantities for the orders placed by 'Vandan' and 'Ivan'.
 6. Find the products and their quantities for the orders placed by client_no 'C00001'
 7. Find the order_no, Client_no, salesman_no where a client has been received by more than one salesman.
 6. To aggregate data using group function and implement the concept of sub-queries.
- Perform following queries based on all 5 tables mentioned above:**
1. Print the description and total quantity sold for each product.
 2. Find the value of each product sold.
 3. Find out the products which have been sold to 'Ivan'.
 4. Find the names of clients who have 'CD Drive'.

5. Find the products and their quantities for the orders placed by 'Vandana' and 'Ivan'
6. Select product_no, total_qty_ordered for each product.
7. Display the order number and day on which clients placed their order.
8. Display the month and date when the order must be delivered.

To implement concept of sub-queries.

1. Find the product_no and description of non moving products.
2. Find the customer name, address, city and pincode for the client who has placed order no "019001".
3. Find the client name who have placed order before the month of may 2006.
4. Find out if product "1.44 Drive" is ordered by only client and print the client_no, name to whom it was soled.
5. Find the name of client who have placed orders worth Rs. 10000 or more.
6. Select the orders placed by "Rahul Desai".
7. Select the name of person who are in Mr.Pradeep's department and who have also worked on inventory control system.
8. Select all the clients and the slaesman in the city of Bombay.
9. Select slaesman name in Bombay who has atleast one client located at Bombay.
10. Select the product_no, description, qty_on-hand, cost_price of non moving items in the product_master tab

7. To implement the concept of views and indexes

1. Create an index on the table client_master, field client_no.
2. Create an index on the sales_order, fields_order_no.
3. Create a composite index on the sales_order_details table for the columns_order_no. and product_no.
4. Create view on salesman_master whose sal_amt is less than 3500.
5. Create a view client_view on client_master and rename the columns as name, add1, add, city, pcode, state respectively.
6. Select the client names from client_view who live in city 'Bombay'
7. Drop the view client_view.

8. To implement concept of PL/SQL

1. WAP in PL/SQL for addition of two numbers.
2. WAP in PL/SQL for addition of 1 to 100 numbers.
3. WAP in PL/SQL to inverse a number, eg. NUMBER 5639 when inverted must be display as output 9365.

9. To implement concept of cursor

Create a explicit cursor which updates the salary of an employee such that,

1. If salary > 10000, then increase the salary by 15%
2. If 5000 < salary < 10000, then increase the salary by 12%
3. Otherwise, increase the salary by 25%

10.To create procedures and functions and triggers in Oracle.

```
DECLARE
    x VARCHAR2(20);
BEGIN
    SELECT RTRIM(TO_CHAR(SYSDATE, 'DAY'), ' ') INTO x from DUAL;
    IF x = 'SUNDAY' THEN
        RAISE_APPLICATION_ERROR (-20001, 'Transaction is not allowed');
    END IF;
END;
```

Solution2.This package has one subprogram; procedure, update_sal.

```
CREATE OR REPLACE PACKAGE emp_pack
IS
    PROCEDURE update_sal (eno IN NUMBER);
END emp_pack;
CREATE OR REPLACE BODY emp_pack
IS
    PROCEDURE update_sal (eno IN NUMBER)
```

IS

x EMP.Empno % type

y EMP. Sal % type

BEGIN

SELECT empno, sal INTO x, y FROM emp WHERE empno=eno;

IF y> 3000 THEN

UPDATE EMP SET Sal= sal*1.1 WHERE Empno =eno;

ELSIF y between 2000 AND 3000 THEN

UPDATE EMP SET Sal= Sal*1.05 WHERE Empno =eno;

ELSE

UPDATE EMP SET Sal= Sal*1.03 WHERE Empno =eno;

END IF;

Calling a Package

Calling a package means actually referencing one of its elements. Following is the method for calling an element from a package.

DECLARE

x VARCHAR2(20);

BEGIN

emp_pack.update_sal (7633);

END;

Total Hours: 30 hours**Mode of Evaluation**

CA1 12	CA2 13	ESE	Total
25	25	50	

Course Code: CS206P	Course Name: Operating System Lab								L	T	P	C
									0	0	2	1
Pre-requisite: Basic concept of C language and data structure												
Course Objectives: This course provides hands-on experience with operating system concepts, focusing on UNIX/Linux environments. After this course student will be able to apply the operating system algorithms to solve the real-life problems.												
Course Outcome: After completion of the course, the student will be able to												
1. Apply knowledge of basic UNIX System calls and Shell programming.												
2. Implement various CPU scheduling algorithms and deadlock handling techniques.												
3. Implement memory management, process synchronization techniques, page replacement techniques, disk scheduling												
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)												
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	3	-	-	1	1	1	1	2
CO2	3	3	1	2	2	-	-	1	2	2	1	2
CO3	3	3	-	2	2	-	-	1	2	2	1	2
List Of Practical's (Indicative & Not Limited To)												
1. Study of hardware and software requirements of different operating systems (UNIX, LINUX, WINDOWS, Android, IOS												
2. Execute various UNIX system calls												
3. Write a basic program on VI Editor in Linux.												
4. Shell Scripting.												
5. Implement FCFS CPU Scheduling Policy												
6. Implement SJF CPU Scheduling Policy												
7. Implement Priority CPU Scheduling Policy												
8. Implement Round Robin CPU Scheduling Policy												
9. Implementation of Banker's algorithm												



10. Implement the solution for Bounded Buffer (producer-consumer) problem using inter process communication techniques-Semaphores			
11. Implement the solutions for Readers-Writers problem using inter process communication technique – Semaphore			
12. Implementation First Fit contiguous allocation technique			
13. Implementation Best Fit contiguous allocation technique			
14. Implementation Worst Fit contiguous allocation technique			
15. Implement file storage allocation technique: -Contiguous (using array)			
16. Implement file storage allocation technique: - Linked –list (using linked-list) -Indirect allocation (indexing)			
17. Comparison of Disk Scheduling Algorithms.			
Total Hours: 30 hours			
Mode of Evaluation			
CA1 12	CA2 13	ESE	Total
25		25	50

Course Code: CS301P	Course Name: Object Oriented Programming using Java Lab								L	T	P	C
									0	0	2	1
Pre-requisite: Basic concept of C language and data structure												
1. To familiarize students with the basic and advance Java Programming Language.												
2. To learn modern tools to develop java-based web applications.												
Course Outcome: After completion of the course, the student will be able to												
1. Perform Java oops concepts on an integrated development environment to solve real world problems.												
2. Solve problems in context of programming code based on collections and new java features.												
3. Develop a solution for case study-based problem using advance java concepts.												
4. Design RESTful Web Services with Spring Boot Test using Spring Framework concepts												
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)												
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	2	2	2	-	-	-	-	-	-	2
CO2	2	3	2	2	2	-	-	-	-	-	-	3
CO3	2	2	2	2	3	-	-	-	-	-	2	3
CO4	2	3	3	3	2	-	-	-	-	-	2	3
List Of Practical's (Indicative & Not Limited To)												
1. WAP to insert 3 numbers from the keyboard and find a greater number among 3 numbers.												
2. WAP to count the total number of odd numbers between 1-100, and display the sum of them.												
3. WAP to Find largest and smallest numbers in an array.												
4. WAP to count the number of charcters in a given string, to reverse the string and check whether it is palindrome or not?												
5. WAP to find out the sum of command line arguments.												
6. WAP to create a class “SimpleObject” and display messages by using the constructor of this class.												
7. WAP to create class Number with only one private instance variable as a double primitive type, include the following methods isZero(), isPositive(), isNegative(), isOdd(), isEven(), isPrime(), isAmstrong() in this class and all above methods should return boolean primitive type like for isPositive() should return “Positive = True”.												
8. WAP to create a Room class, the attributes of this class are roomno, roomtype, roomarea and ACmachine. In this class the member functions are setData and displayData. Use member function to set data and display that data using displayData() method.												
9. WAP to create a class named Shape and create three subclasses Circle, Triangle and Square, each class has two-member functions named draw () and erase (). Implement this concept using polymorphism.												
10. WAP to create a class Employee with a method called calculateSalary(). Create two subclasses Manager and Programmer. In each subclass, override the calculateSalary() method to calculate and return the salary based on their specific roles.												

11. WAP to implement the concept of method overloading and method overriding. And Abstract Class.
12. WAP to Create a class Employee having members as follows:
private int empNo
private String empName
private int empBasic
Parameterized constructor to initialize members.
Getter methods for all instance variables
13. Create a class WriteEmployee having a main method. Ask users to enter details of an employee and set them in an Employee object. Store details of this object in a file emp.txt. Read employee details from the file and display those details.
14. WAP to create a text file in the path c:/Java/abc.txt and check whether that file exists or not. Using the commands exists(), isDirectory(), isFile(), getName() and getAbsolutePath().
15. WAP to Implement three classes: Storage, Counter and Printer.
The Storage class should store an integer.
The Counter class should create a thread and start counting from 0 (0,1,2, 3 ...) and store each value in the Storage class.
The Printer class should create a thread that keeps reading the value in the Storage class and printing it.
Write a program that creates an instance of the Storage class and set up a Counter and Printer object to operate on it.
16. WAP to Create a class MyThread derived from Thread class and override the run method. Create a class ThreadDemo having a main method. Create 2 objects of MyThread class and observe the behavior of threads.
17. WAP to Modify the above to create MyThread class by implementing Runnable interface and observe the behavior of threads.
18. WAP to Assign different priorities to the 2 threads and observe the behaviour.
19. WAP to create a producer-consumer scenario using the wait() and notify() methods for thread synchronization.
20. WAP to implement deadlock in Java (Content Beyond Syllabus).
21. WAP to implement the following new features in Java.
Functional Interface, Lambda Expression: Write a Java program to implement a lambda expression to check if a given string is empty. Method References: Default and Static Method in Interface Inner Class
22. WAP to implement different types of Annotations in JAVA.
23. WAP to filter data by using streams.
24. WAP to Traversing the array elements and to sum the elements using For-each loop.
25. WAP to implement Base64 Encoding and Decoding.
26. WAP to implement Local Variable Type Inference.
27. WAP to implement Sealed Class.
28. WAP to implement Text Blocks and Records.
29. WAP to iterate a linked list in reverse order.
30. WAP to append the specified element to the end of a hash set.
31. WAP to add all the elements of a specified to another tree set.
32. WAP to count the number of key-value (size) mappings in a map.
33. WAP to search for a value in a Tree Map.
34. WAP to Demonstrate Iterator
35. WAP to convert an Iterable to Collection in Java
36. WAP to Create industry-oriented applications using Spring Framework.
37. WAP to test RESTful web services using Spring Boot.

Total Hours: 30 hours**Mode of Evaluation**

CA1 12	CA2 13	ESE	Total
25	25	50	

Course Code: CS401P	Course Name: Design & Analysis of Algorithms Lab	L	T	P	C
		0	0	2	1
Pre-requisite: The course requires a background in mathematics and strong programming skills .					
Course Objectives:					
1. Implement and analyse Greedy, DP, Backtracking, and Branch & Bound techniques for efficient problem-solving.					



2. Evaluate the performance and complexity of sorting, graph, and optimization algorithms through hands-on programming.

Course Outcome: After completion of the course, the student will be able to

1. Implement and analyse Greedy, DP, Backtracking, and Branch & Bound for efficient problem-solving.
2. Evaluate algorithm performance and complexity through hands-on programming in sorting, graphs, and optimization.
3. Develop problem-solving skills by implementing Knapsack, Matrix Chain Multiplication, Graph Colouring, and TSP.
4. Understand NP-Completeness and Approximation Algorithms for tackling computationally hard problems.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	2	-	-	-	-	-	-	2
CO2	3	3	3	3	3	-	-	-	-	-	-	2
CO3	3	3	3	3	3	-	-	-	-	-	-	2
CO4	3	3	3	3	2	-	-	-	-	-	-	2

List Of Practical's (Indicative & Not Limited To)

1. Implement and analyze **Merge Sort** using recurrence relations.
2. Implement **Quick Sort** and **Randomized Quick Sort** and compare efficiency.
3. Implement **Shell Sort** and analyze its time complexity.
4. Implement **Counting Sort, Radix Sort, and Bucket Sort**, and compare their performance.
5. Solve the Fractional Knapsack Problem using the Greedy approach.
6. Implement the Activity Selection Problem using Greedy Strategy.
7. Solve Task Scheduling with Deadline and Penalty using Greedy Strategy.
8. Implement Kruskal's Algorithm to find the Minimum Spanning Tree (MST).
9. Implement Prim's Algorithm to find the MST.
10. Implement 0-1 Knapsack Problem using Dynamic Programming.
11. Solve the Optimal Binary Search Tree (OBST) problem.
12. Implement Matrix Chain Multiplication using Dynamic Programming.
13. Solve the Longest Common Subsequence (LCS) Problem.
14. Implement the All-Pairs Shortest Path Algorithm (Floyd-Warshall).
15. Given a set of coin denominations and a target amount N, determine the **minimum number of coins** needed and the **total number of ways** to make N using the given denominations.
16. Implement the N-Queens Problem using Backtracking.
17. Solve the Subset Sum Problem using Backtracking.
18. Implement Graph Coloring using Backtracking.
19. Solve the Hamiltonian Cycle Problem using Backtracking.
20. Given an N×N chessboard, the Knight's Tour problem requires finding a sequence of moves where a knight visits **every square exactly once** without repetition. The knight moves in an **L-shape** (two squares in one direction and one perpendicular).
21. Implement Traveling Salesman Problem (TSP) using Branch and Bound.
22. Implement the Naïve String Matching Algorithm.
23. Implement Rabin-Karp String Matching Algorithm.
24. Implement Knuth-Morris-Pratt (KMP) String Matching Algorithm.
25. Solve the Vertex Cover Problem using Approximation Algorithms.

Total Hours **30 hours**

Mode of Evaluation

CA1 12	CA2 13	ESE	Total
25	25	25	50

Course Code: IT302P	Course Name: Computer Networks Lab	L	T	P	C
		0	0	2	1
Pre-requisite: Computer Fundamentals and C programming					
Course Objectives:					



The objective of this course is to provide students with a theoretical and practical base in computer networks issues and prepare the students for easy transfer from academia into practical life.

Course Outcome: After completion of the course, the student will be able to

1. Understand the fundamental concepts of computer networking and Network topologies.
2. Analyze different types of network devices and simple computer networks.
3. Implement the basic network commands and use techniques, skills, and modern networking tools necessary for engineering practice.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	-	-	2	-	-	-	-	-	1	2
CO2	3	2	-	-	2	-	-	-	-	-	1	2
CO3	3	3	2	2	2	-	-	-	-	2	1	2

List Of Practical's (Indicative & Not Limited To)

1. To study of LAN using different Network topologies and different network connecting devices like Hub, Switch, Router etc.
2. To study and learn the handling of networking hardware like: NIC card, RJ-45 connector, crimping tool, CAT-6 cable and Do the following Cabling works in a network (T568A and T568B) a) Cable Crimping b) Standard Cabling and c) Cross Cabling d) IO connector crimping e) Testing the crimped cable using a cable tester.
3. Running and using services/commands like ifconfig, ping, traceroute, nslookup, telnet, ftp, etc.
4. Establish a physical peer to peer network connection using two or more systems using network interface card, network cables, Switch and Router in a LAN.
5. To install the network packet analysis tool Wireshark and also capture and analyze the local traffic.
6. Simulate and Configure a Network topology using packet tracer software.
7. Implement the data link layer framing methods such as character count, character stuffing and bit stuffing.
8. Implement the error correcting code Cyclic Redundancy Check (CRC) of data link layer using various polynomials like CRC-CRC 12, CRC 16 and CRC CCIPP.
9. Implement a program to determine if the IP address is in Class A, B, C, D, or E and translate dotted decimal IP address into 32-bit address.
10. Implement Dijkstra algorithm to compute the shortest path through a given graph.

Total Hours **30 hours**

Mode of Evaluation

CA1 12	CA2 13	ESE	Total
25		25	50

Course Code: IT205P	Course Name: Data Engineering Architecture							L	T	P	C	
								0	0	4	2	
Course Objectives:												
The primary objective of this lab is to provide learners with extensive hands-on experience in designing, ingesting, transforming, storing, operating, securing, and governing data on AWS through guided labs and practical exercises. The course emphasizes building and operating scalable data pipelines, implementing optimized storage architectures and applying compliance-driven data governance using real-world AWS services aligned with the AWS Data Engineer Associate certification.												
Course Outcome: After completion of the course, the student will be able to												
1. Understanding cloud-based data engineering principles, ingestion patterns, and data lifecycle stages. 2. Converting raw sources into structured, analytics-ready datasets through consistent workflows. 3. Designing efficient, scalable, and cost-aware data storage with appropriate structures and tiers. 4. Monitoring pipelines, validating data, tuning performance, and ensuring reliability. 5. Protecting data with access controls, encryption, auditing, and compliance-based policies.												
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)												
CO-PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12

Mapping												
CO1	3	2	1	-	2	-	-	-	-	-	-	2
CO2	3	3	2	2	3	-	-	-	-	-	-	2
CO3	3	2	3	-	3	-	2	-	-	-	2	1
CO4	2	3	2	3	3	-	-	-	-	-	2	2
CO5	2	2	2	-	2	3	-	3	-	-	-	2

List Of Practical's (Indicative & Not Limited To)

Lab 1: Designing a Layered Data Pipeline with Controlled Data Zones

Design a layered data pipeline that ingests operational data into S3 and progressively refines it through well-defined data zones. IAM roles and bucket policies enforce zone-level access control.

Lab 2: Event-Oriented Pipeline Design with Decoupled Components

Build an event-driven pipeline where data arrival automatically triggers validation and metadata updates.

Lab 3: Batch ETL Pipeline with Schema Evolution Handling

Implement a batch ETL pipeline that processes semi-structured data with evolving schemas. The pipeline supports backward compatibility while converting data into columnar formats. Emphasis is placed on idempotency, schema evolution, and reliable transformations.

Lab 4: Streaming-to-Batch Hybrid Pipeline Design

implements a hybrid pipeline that ingests streaming data and periodically consolidates it into batch datasets for analytics. Learners manage checkpoints, micro-batching, and late-arriving data.

Lab 5: Performance-Optimized Data Lake Design

Optimize a data lake by applying partitioning strategies, converting datasets to Parquet, and tuning query performance. Performance improvements are measured using Athena execution metrics.

Lab 6: Cost-Aware Storage Tiering and Lifecycle Automation

Implement automated data lifecycle policies to transition data across storage tiers. Dataset tagging and retention strategies are applied to balance long-term cost and accessibility.

Lab 7: Operationalizing a Data Pipeline with Monitoring and Alerts

Operationalize a multi-step pipeline by configuring logging, alarms, and retry mechanisms. The lab emphasizes SLA-driven monitoring and performance tuning.

Lab 8: Data Quality Enforcement within Processing Pipelines

Integrate data quality checks directly into ETL workflows. Invalid records are isolated into a quarantine zone for further analysis, reinforcing quality-by-design principles.

Lab 9: Fine-Grained Permissions Management for a Data Lake

Design role-based access control for ingestion, processing, analytics, and administration roles. Policies enforce least-privilege access and auditability.

Lab 10: Governance-Driven Pipeline Design with Auditing

Embed governance requirements into pipeline design by enforcing tagging standards, auditing data access, and validating end-to-end data lineage and traceability.

Total Hours **30 hours**

Mode of Evaluation

CA		ESE	Total
CA1	CA2		
25	25	50	100

CA1: 25 Marks

- CA-1 will be conducted online using platforms like CodeTantra. The exam will include use-case-based questions focusing on practical understanding and problem-solving skills.

CA2: 25 Marks

- CA2 on online mode projects based on given case studies. **OR**
- Quiz 100% based on Data Engineering Architecture using platforms like CodeTantra.

ESE (50 Marks):

- Certification Exam- Data Engineering Architecture Certification.(Marks mapped according to the certification score.)